

2023 Sustainability Report

Improving Everyday Life in a Lower Carbon World



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Senior Management Letter

Magris Performance Materials is pleased to present our second annual sustainability report.

After publishing our first sustainability report last year and committing to the United Nations Global Compact, we continued to build on our sustainability foundations in 2023 through our efforts to minimize our potential impact on others and the environment and live up to our vision—supplying materials that enhance the performance and sustainability of critical products for a lower carbon world.

Our sustainability gains in 2023 included updating all our sustainability-related policies based on recommendations from our 2022 third-party assessment, standardizing how we track Scope 1 and Scope 2 greenhouse gas (GHG) emissions, and documenting our main GHG sources as we work toward our goal of establishing a net-zero carbon emissions roadmap. We continued to recycle nearly 70 percent of the water we consume in our operations. We also began creating a formal framework for supply chain partner compliance reviews and audits and completed diversity, equity and inclusion training for all managers.

We are particularly proud of our achievements in Workplace Health, Safety, and Wellbeing—one of our 3 priority sustainability topics. Magris experienced no fatalities or life-altering events and a 28% decrease in our total recordable injury rate

in 2023. We increased our focus on in-field presence and high-severity workplace risks, enhanced our risk management framework, and implemented cybersecurity and IT improvements after our 2022 360° systems review.

Guided by our core values—Respect, Integrity, Safety, and Excellence—we are building Magris into a leading materials business focused on creating sustainable, long-term value. In 2024, we will continue to improve our health and safety culture, identify opportunities to make our workplaces more welcoming to all, further our road map for achieving net-zero carbon emissions by 2050, and establish a permanent solution for management of effluent discharge at our Canadian talc operations.

We invite you to explore this report to better understand our progress over the last year and our commitment to sustainable operations that create long-term value.

Yours truly,

Aaron Regent
Chair & Chief Executive Officer

Matthew Fenton
President & Chief Financial Officer



Aaron Regent
Chair & Chief Executive Officer



Matthew Fenton
President & Chief Financial Officer



About This Report

In this Sustainability Report, we discuss our business, vision, values, and performance related to relevant sustainability topics. Our assessment and identification of relevant topics is informed by both global and industry sustainability-related trends, frameworks, expectations, and best practices.

How we disclose our performance for relevant topics reflects the [Sustainability Accounting Standards Board \(SASB\)](#) Metals and Mining Standard (MMS). Where MMS indicators do not exist for these topics, we employ [Global Reporting Initiative \(GRI\)](#) standards. While we currently reflect the MMS, we continue to evaluate the applicability of other standards to us as the sustainability disclosure landscape continues to evolve.

The Magris Performance Materials Inc. (MPM) Board of Directors has reviewed and approved this Report. The Magris senior management teams have reviewed the contents of this Report to confirm that it includes all information and sustainability topics that we believe are material to us, and our technical and subject matter experts have reviewed all included data to confirm its accuracy.

All information in this Report is current as of publication. All monetary figures in this Report are in US Dollars unless otherwise indicated, “C\$” = Canadian dollars, “t” = metric tonne, and “CO₂e” = carbon dioxide equivalent.

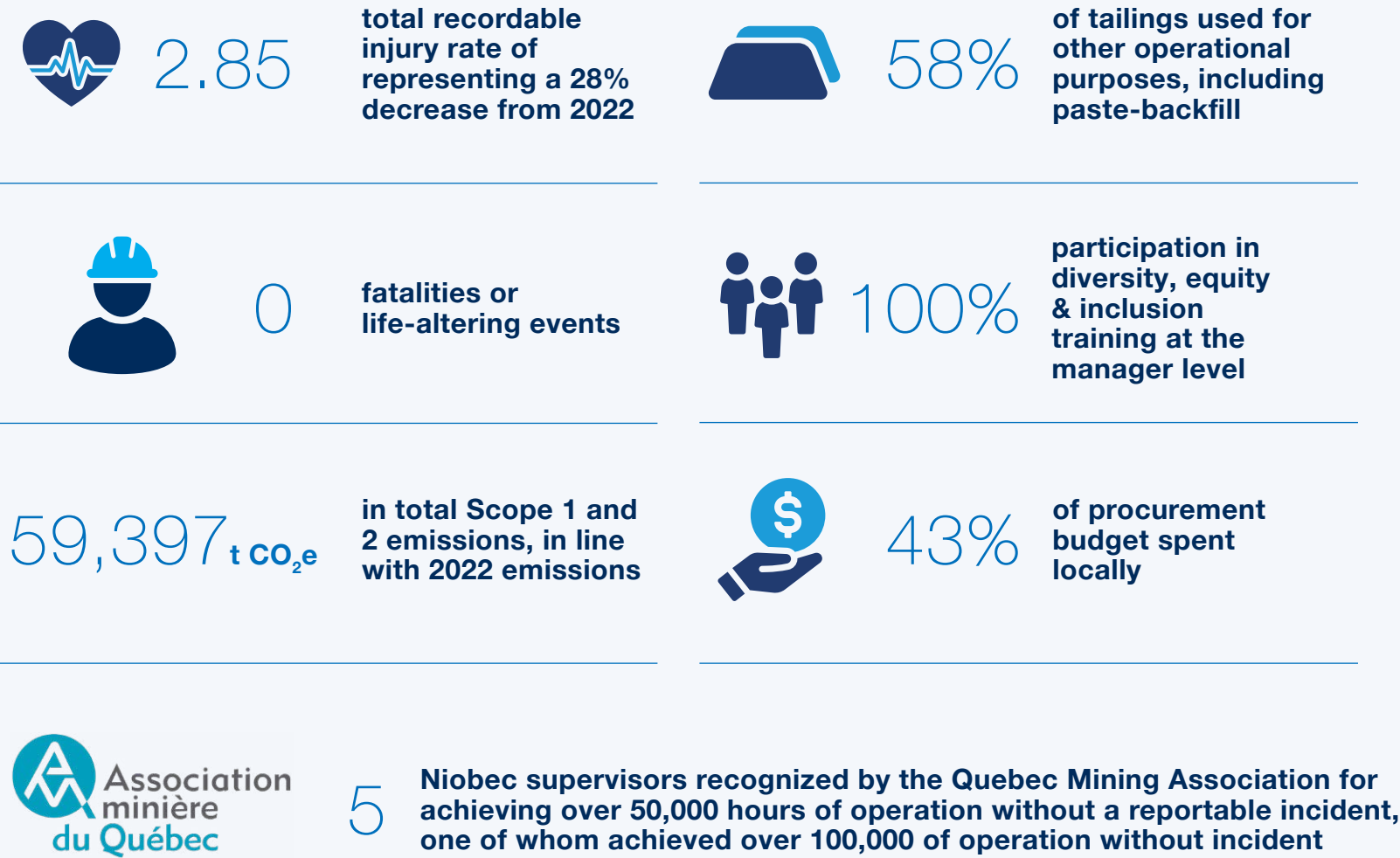
Key Terms

In this Report, “Magris”, “us”, “we”, and “our” refers collectively to MPM and its subsidiaries that supply our products to customers – Niobec Inc. (Niobec), Niobec GmbH (GmbH), Magris Talc USA, Inc. (MT) and Magris Talc Canada Inc. (MTC). When we refer to Magris Talc, we mean MT and MTC.

Cautionary Statement

This Report contains statements about our business activities, products, markets, operational results, and financial condition that we believe are accurate and complete as of the date of publication. Some of these statements may be “forward looking” because we have used what we know and expect today to make a statement about the future. Forward-looking statements usually include words such as may, expect, anticipate, believe or similar words. We believe the expectations reflected in any forward-looking statements in this Report are reasonable. Actual events and results could be substantially different because of the risks and uncertainties associated with our business or events that happen after the date of this Report. For more information about Magris, please visit us at www.magrispm.com. We welcome any questions or comments on this Report. Please reach out to us at: info@magrispm.com

2023 Highlights



About Magris

Our Operations

Magris is a privately-owned, North American-based performance materials producer supplying manufacturers with key inputs for their essential products.

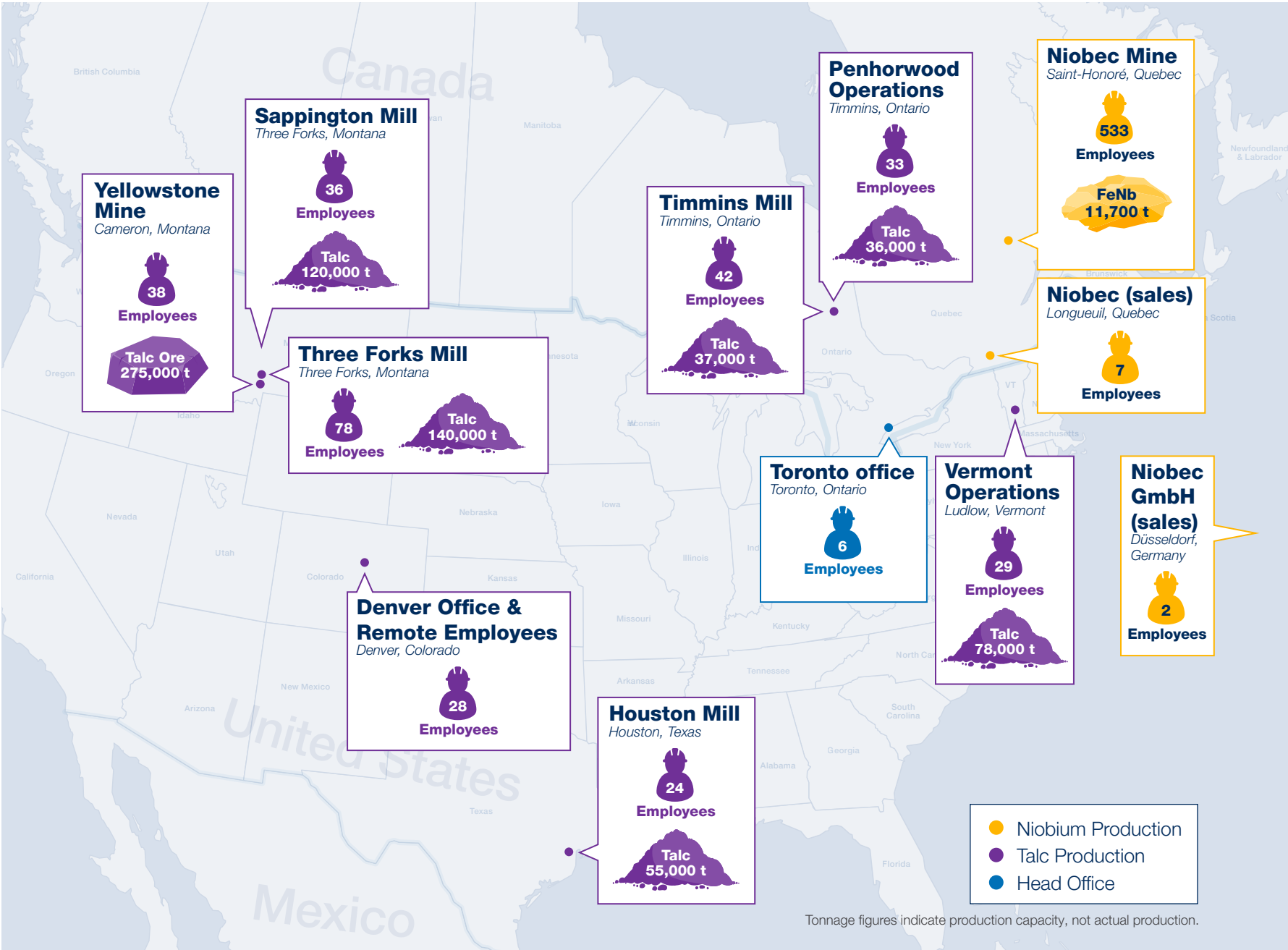
The critical materials that we produce, niobium and talc, are used in a variety of industrial applications to enhance performance and improve sustainability in everyday products. Manufactured from a portfolio of world-class, vertically integrated mining and manufacturing facilities in Canada and the United States, our products are sold globally. We have owned our niobium business since 2015 and our talc business since 2021. In 2023, we had revenues in excess of \$450 million. We supply approximately 50% of North America’s talc market and 10% of the global niobium market from our low-environmental footprint facilities. For more information about our products and operations, please see the “[More About Magris](#)” appendix to this Report.

Our vision guides us in what we do:

We supply materials that enhance performance and sustainability in critical products that improve everyday life in a lower carbon world. We operate responsibly—respecting our employees, communities, and the environment.

Our corporate head office is located in Toronto, Ontario while Magris Talc has a regional office in Denver, Colorado. Niobec owns and operates a niobium mining, processing, and metallurgical complex in the Saguenay region of Quebec, with niobium sales offices in Longueuil, Quebec (Niobec) and Duesseldorf, Germany (GmbH). Our Magris Talc operations are located in Montana, Vermont, Texas, and Ontario. We also maintain and are responsible for several non-operating, legacy facilities in California, Montana, and Vermont stemming from the 2021 acquisition of our talc business. In addition, we own a rare earths elements deposit in the vicinity of our Niobec facility.

Magris employs over 850 people across all our locations, with the vast majority of them being local community members.



How We Improve Everyday Life

Our products are **performance materials**, meaning they provide more sustainable and lower-carbon solutions to a range of industries, including manufacturing, agriculture, and transportation.

Improving everyday life through:

More information about our products can be found in the "[More About Magris](#)" appendix of this Report.

Niobium



Where our Niobium is used



Automobiles

Niobium results in stronger and lighter auto components with improved performance and better fuel efficiency, with up to **10% less** weight in vehicles.

Potential future applications include enhanced electric vehicle (EV) battery performance (significantly reduced charging times/higher capacity), greater life (more charging cycles available) and safety.



Infrastructure and Construction

Niobium contributes to dematerialization, making more goods with fewer resources. When used in construction steels, niobium typically results in an average of **22% less material input**.

Niobium in pipelines can **reduce total materials used by up to 24%** to transport the same volume.

Talc



Where our Talc is used



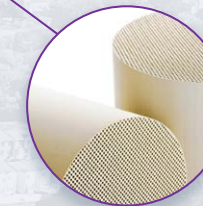
Polymers

Talc used in polymers in plastic auto parts adds tensile strength, rigidity and impact resistance, lightweighting vehicles and increasing fuel efficiency.



Rubber

Talc provides dimensional and thermal stability, lubricity, dusting, and reinforcement properties to hoses, window seals, and tires.



Specialties

Talc in catalytic converters reduces harmful exhaust gases and particulate emissions. Talc in ceramics adds thermal stability, crack and shrink resistance, and stiffness.



Agriculture

Talc-based solutions in agriculture can provide natural alternatives to chemicals that protect plants from insects and sunburn.



Construction

Talc enhances crack and shrinking resistance, corrosion resistance and barrier properties in adhesives, caulks, sealants and asphalt roofing.



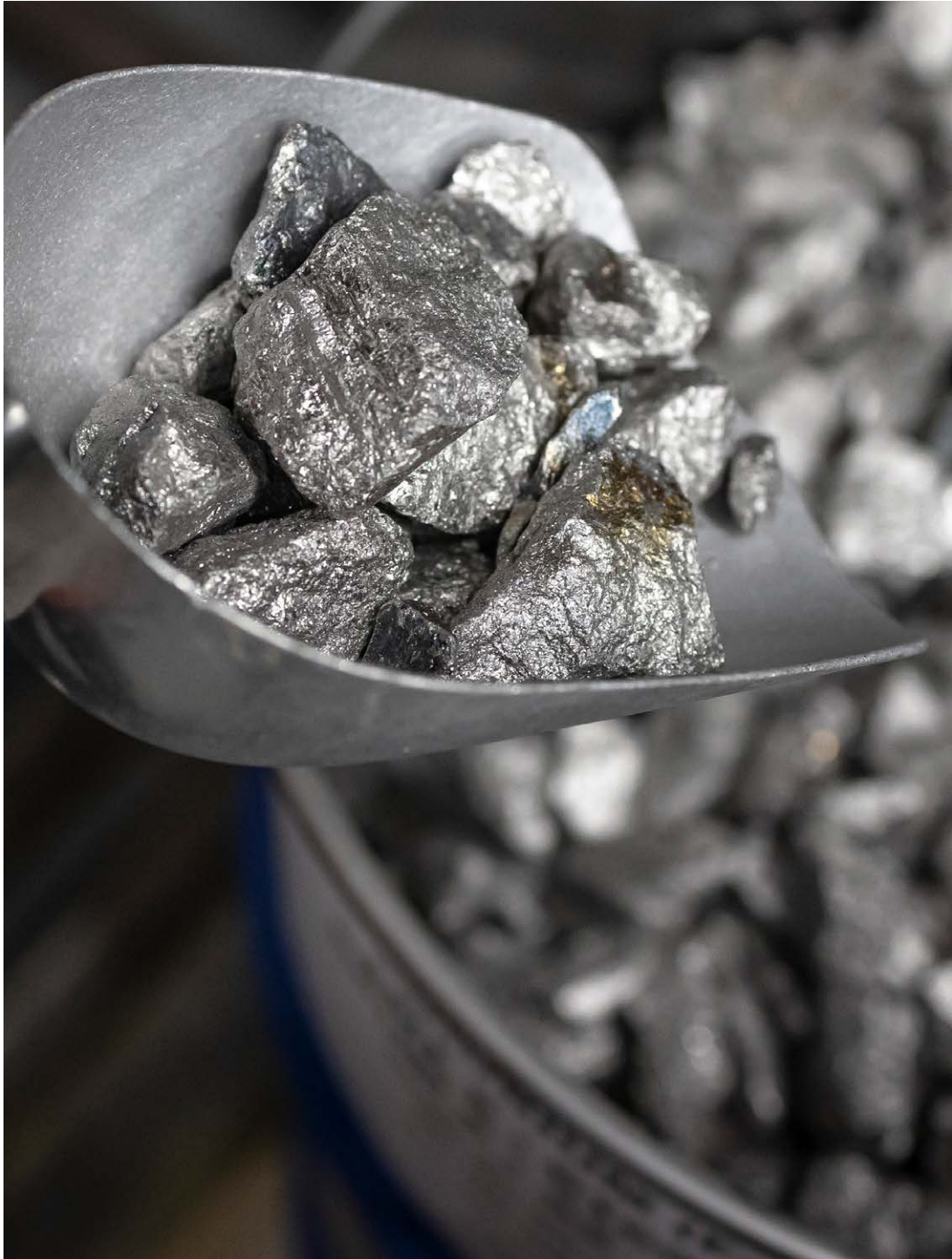
Paints & Coatings

Talc adds corrosion resistant properties to protective gels and marine gels, gel coats, primers, cured-in-place pipe and architectural coatings.



Paper

Talc in paper production displaces harmful chemical-based alternatives.



How Does Our Niobium Advance a Low Carbon Future?

Climate change is one of the most pressing issues facing humanity, and the risks associated with a changing climate – risks to the planet, people, and business – can no longer be ignored. We must do our part to address the challenge of climate change and reduce our impacts, implement mitigation and adaptation activities, and provide products that contribute to a lower carbon world.

In 2022, we commissioned an external life-cycle assessment (LCA) study that evaluated the carbon footprint of our ferroniobium (FeNb) product and identified the enabled benefits for its various downstream applications.*

Some of the enabled benefits identified by the study include niobium applications in construction, automotive, and pipelines that contribute to reductions in GHG emissions:

Construction: Using our niobium in high-strength low-alloy (HSLA) steel in construction applications typically results in an average of 22%, and up to 60%, less material input compared to using conventional steel.

Automotive: Using niobium in automotive components typically leads to 17-30% less steel input compared to using conventional steel. Less weight results in greater fuel efficiency and lower emissions over the life of a vehicle.

Pipelines: Using HSLA steel with higher niobium content allows for the construction of pipelines with twice the capacity, resulting in an average reduction of 24% in steel input compared to standard pipelines.

The study found that the carbon footprint of our FeNb production is 15.96kg CO₂e/kg (based on a three-year average from 2019-2021) which equates to total annual GHG emissions of 154,000 tCO₂e from our FeNb production.

If we scale the niobium-derived benefits using Niobec’s average annual sales volume, the potential annual GHG emissions avoided are 9 million t CO₂e, with certain scenarios suggesting a top end of 35 million t CO₂e in avoided emissions. The 9 million t CO₂e of avoided emissions represents 58 times the annual GHG emissions we generate in producing our FeNb. This ignores other benefits resulting from the use of fewer resources and materials that were not assessed as part of the study.

* The study included the years 2019, 2020, and 2021, and was performed in accordance with the Product Life Cycle Accounting and Reporting Standard developed by the Greenhouse Gas Protocol. The study considered mining operations as well as energy and material consumption used in concentrating, converting, and processing niobium ore to ferroniobium, and included an analysis of the enabled carbon benefits associated with downstream niobium use in automotive, construction, and pipeline applications. While the study is based on data from 2019-2021, there has been no change in our production methods for niobium since then, and so we continue to believe that the study and its conclusions remain valid.

Sustainability at Magris

We define sustainability as our ability to manage the environmental and social impacts of our activities so we create value in the long-term—meeting the needs of the present without compromising the ability of others and future generations to meet their needs.

Magris was founded as a business enterprise that prioritizes sustainability to create value in a responsible manner. Our commitment to sustainability underlies our focus on our long-term business viability, our operational and financial performance, and the potential impacts we have on others and the environment. As a global business, we are also subject to and must manage sustainability-related operational, financial, and reputational risks as well as evolving societal concerns and expectations connected to sustainability.

Our Core Values

Our business is built on four core values: Respect, Integrity, Safety, and Excellence. These values guide us to openly engage with those affected by our actions and to consider their diverse and sometimes conflicting perspectives before we make decisions.



Respect

We respect each other, the natural world around us, and all those with whom we interact or who are affected by our activities



Integrity

We are honest and open, and act in a way we can be proud of



Safety

We believe that everyone should go home safe and healthy, every day



Excellence

We strive to be excellent in everything we do and embrace teamwork and creativity

We are also guided by leading international standards, practices and industry organizations, including the Mining Association of Canada's Towards Sustainable Mining (TSM) standard. This standard is the basis of a globally recognized sustainability program that supports mining to manage key environmental and social challenges. As a member of the Quebec Mining Association, Niobec publishes annual performance results against the TSM framework. You can click [here](#) to view Niobec's most recent TSM Results Report.

We became a voluntary participant in the United Nations Global Compact (UNGC) in 2022 as a demonstration of our commitment to sustainability in our businesses and will file our first Communications on Progress in 2024. You can read about the UNGC's Ten Principles at www.unglobal.com/what-is-gc/mission/principles.



Sustainability Governance and Management

Our governance of sustainability matters occurs at multiple levels with an emphasis on communication, transparency, and responsiveness between levels:

Governance

Magris Performance Materials Board of Directors

The MPM Board of Directors (MPM BoD) has overall responsibility to supervise the management of the business and affairs of MPM. As part of its responsibility, it oversees, reviews, and approves the sustainability strategy and sustainability-related policies of MPM. It also reviews and authorizes funding appropriations for sustainability-related operating and capital expenditures as part of the annual consolidated budgeting and planning process or where required under Magris authorization policies in connection with any discrete authorization request from MPM executive management. The MPM BoD does not delegate responsibility for sustainability-related matters to a committee, and each director is directly involved in overseeing the sustainability-related aspects of MPM’s business. The MPM BoD receives updates on sustainability-related matters, performance and initiatives at quarterly board meetings or through written updates and briefings from the MPM executive management team. These matters are reviewed and discussed with the MPM executive management team and by Niobec and Magris Talc senior management team members. Additional MPM BoD meetings are held as needed.

Operating Subsidiary Boards

The board of each operating subsidiary (Magris Talc and Niobec) comprised of management of the subsidiary and MPM (and in the case of Niobec, certain MPM directors as well), also receives information periodically and as needed on sustainability-related matters that pertain to the business of the subsidiary. Each board reviews and authorizes associated sustainability-related expenditures and actions separately from the MPM BoD.

Governance



Management

MPM Executive Management

The MPM executive management team (Chair & CEO, President & CFO, Vice-President, Operations, and Vice-President, Commercial and Corporate Affairs) formulates the sustainability strategy and Magris-wide policies and presents them to the MPM BoD for review, discussion, and approval. It also makes authorization requests to the MPM BoD for significant sustainability-related expenditures and actions. The MPM executive management team also guides the Niobec and Magris Talc management teams regarding prioritization and resource allocation for sustainability-related matters.

Niobec and Magris Talc Senior Management Teams

The Niobec and Magris Talc senior management teams provide input to the MPM executive management team on sustainability strategy, Magris-wide policies, actions, and related operating and capital expenditure requirements. Each team, working with its operating personnel and subject matter experts (SMEs), both internal and external, has frontline responsibility for identifying and assessing sustainability-related risks and opportunities that arise in its business, and for proposing actions to the MPM executive team to manage such risks or act on such opportunities. Each team is also responsible for the implementation of sustainability-related initiatives and actions in its business at the operational level, in line with Magris-wide policies, commitments and plans.

Sustainability Management Committee

This committee is comprised of members of the MPM executive management team, senior Niobec and Magris Talc management, and operations managers with sustainability-related responsibilities. The committee meets monthly, and more frequently if necessary, to discuss and review sustainability-related risks and opportunities, review and revise sustainability priorities, set annual and ongoing sustainability-related objectives and targets, and recommend plans of action to achieve and assess progress made towards sustainability-related objectives.

Members of the MPM executive management team are also in frequent, if not daily, contact with Niobec and Magris Talc management members concerning ongoing sustainability-related matters or issues that may arise between Sustainability Management Committee meetings.

Our Approach to Sustainability

The complexities of our business and the world we operate in mean that addressing sustainability is crucial for our long-term viability. By identifying, measuring, and evaluating sustainability-related risks and opportunities, we can create value, enhance operational and financial performance, and grow our business. We view sustainability-related risks and opportunities as part of our overall risk management strategy. We recognize that some sustainability topics are inherently subjective and may lack high-quality data. In such cases, we take a pragmatic approach, balancing sustainability implications with our understanding to ensure appropriate action.

The 2021 acquisition of our talc business presented us with an opportunity to review the separate, existing management systems and tools at Niobec and Magris Talc. Work is ongoing to harmonize these systems and tools for risks and opportunities common to both businesses, including those associated with sustainability.

Policies

We developed the Magris Sustainability Policy to define what sustainability means to us and to guide our daily efforts in managing sustainability-related risks. This policy helps us positively impact the environment, people, and communities, and enhance the long-term viability of our business. You can view the Sustainability Policy and our other corporate policies that relate to the topics discussed in this Report at www.magrispm.com/sustainability.

To strengthen our sustainability management framework, we updated our corporate policies for our priority sustainability topics in 2023. We implemented these updated policies across Magris in the first quarter of 2024. Each policy outlines the principles and standards guiding our commitments and actions for its subject matter. These principles are incorporated into the policies, standards, processes, and procedures used by each operating subsidiary and corporate function to manage risks in daily activities. Additionally, we developed specific policies on human rights, privacy, improper payments (anti-corruption), sanctions, and anti-trust and competition.

Focusing on What Matters Most

In 2022, we conducted a review, with the assistance of a third-party consultant, which identified and assessed the sustainability topics that were most relevant to our businesses. This review was informed by global and industry sustainability trends and frameworks, expectations, and best practices in the context of the sustainability-related aspects relevant to our business, where our management of these aspects could significantly affect the long-term viability of our business. We identified eight topics as a result of this review that form the basis of this Report. While all of these topics are important to our business, we are especially focused on Workplace Health, Safety & Well-Being, GHG Emissions & Climate Change, and Water & Effluent Management since we believe that these topics have the most significant potential impact on our long-term viability.

We periodically review and reprioritize sustainability-related aspects of our business as a result of the progress we make in achieving our sustainability objectives, the emergence of new issues, and evolving regulatory requirements.

Sustainability Measurement and Reporting

A comprehensive and transparent internal reporting system is essential for assessing, understanding, and addressing sustainability-related risks and opportunities within our business.

We internally track and record our progress toward specific objectives and activities for each sustainability topic using measurable, time-bound, and standardized key performance indicators (KPIs). Depending on the specific KPI, the frequency of internal reporting can be on a weekly, monthly quarterly or annual basis (or any combination of them). These KPIs allow us to continuously evaluate our current performance against targets and past results. We also report these and other KPIs externally, either as mandated by applicable laws or regulations or voluntarily as in the case of this Report.

Niobec and Magris Talc include their KPIs in the regular internal reports they produce. These reports are reviewed and discussed with management and the relevant boards of directors, including the MPM BoD.

Additionally, we are aligning KPIs and tracking tools between Niobec and Magris Talc for common activities or risks. For activities or risks unique to each business, industry-standard KPIs and tracking tools are utilized.





Our Approach to Risk Management

Historically, our facilities focused on identifying operational risks and putting in place systems, procedures, and processes to manage them on an individual basis. While this approach allowed for appropriate risk management practices relevant to a facility, it lacked a cohesive risk management framework with standardized “risk language” and methods for collection, analysis, and reporting of risk-related data and effective sharing of management best practices, across the organization.

Integration

In 2022, after acquiring our talc business, we started work to establish a unified Enterprise Risk Management (ERM) framework. However, over the course of this initiative, we concluded that fully integrating each business’s ERM framework was not optimal due to the unique risks and challenges that Magris Talc and Niobec each face. Instead, in 2023, we focused on harmonizing the separate risk management systems and tools of our two distinct businesses. This framework aims to ensure consistency across all sites by standardizing data collection and analysis methods and adopting a common language and terminology for identifying, tracking, and managing risks and opportunities shared among the businesses. At the same time, each business retains its historical management systems and tools for its unique risks and opportunities.

Our current efforts in harmonization are focus on areas where we perceive sufficient common ground between Niobec and Magris Talc. We believe that greater alignment in these areas will yield tangible benefits. Specifically, our focus is on Workplace Health, Safety & Well-being, GHG Emissions & Climate Change, Workplace Culture, Business Ethics & Transparency, and Responsible Supply Chain.

Responsibilities and Roles

The approach that Magris takes to risk management can best be summarized as “top down, bottom up”.

The MPM executive team drives the formulation of Magris’ sustainability vision and strategy and setting and prioritizing of annual and other objectives for each business and Magris as a whole. For risks and opportunities specific to one business, the MPM executive management team works with the operations management, marketing, and other functional teams of that business to identify and assess them within their areas of responsibility.

The operational and functional teams play a critical role in identifying and assessing risks and opportunities within their business, promptly informing the MPM executive management team and providing recommendations to address the risks and execute on the opportunities. External consultants are engaged where necessary, to support the business’s internal SMEs in comprehensively understanding issues, evaluating options, and determining the appropriate actions to address the risk or act on the opportunity.

If a risk or opportunity extends beyond the operations teams’ responsibility or impacts Magris as a whole, the MPM executive management team is responsible for identifying, evaluating, and determining appropriate actions in consultation with the operations and other teams. This may include efforts such as creating and implementing the Code of Conduct or determining actions for compliance with anti-corruption or sanctions legislation.

Given that certain risks cannot be completely eliminated and opportunities must align with our long-term sustainability vision while also demonstrating commercial viability, we thoroughly evaluate all alternatives through trade-off or other studies. After assessing the alternatives, we proceed with the option that we believe is most appropriate in the circumstances. Associated expenditures to address the risk or opportunity are authorized at the appropriate level either as a unique item or as part of the annual budget and planning process, depending on their nature.

Day-to-Day Operations Management

We manage sustainability-related risks that are associated with our day-to-day operations through the following actions:

- Adhere to site-operating standards that meet or exceed our compliance obligations under applicable laws and regulations and meet the concerns of or our commitments to our stakeholders;
- Set annual and ongoing sustainability-related targets and objectives;
- Track and monitor our performance of our sustainability-related targets and objectives, assess and report our performance, and take action to address non-performance where needed;
- Implement and maintain management practices based on industry recognized systems and standards;
- Monitor and enhance our sustainability-related procedures and practices through regular inspections, audits, reviews, investigations, corrective actions, shared learnings, review of best practices, and behaviour-based processes that lead to continuous improvement;
- Allocate the appropriate resources to fulfill our commitments;
- Raise awareness and encourage, train, and empower our employees and contractors to meet our commitments;
- Use sustainability as a driver to improve our operational and financial performance;
- Define and communicate our environmental and social performance objectives clearly and transparently and report on our performance annually;
- Review and assess our progress towards achieving our environment and social commitments regularly at all levels of our organization; and
- Avoid doing business or maintaining any other relationships with those whose actions are inconsistent with our values and environment and social objectives.



Our Approach to Acquisitions

Magris regularly evaluates potential acquisitions. We consider sustainability-related risks and opportunities in our M&A activities in a similar manner to how we consider such risks and opportunities in our own business.

We conduct due diligence on targets to identify and assess sustainability-related potential risks and opportunities, analyze how we can best manage the risks and act on the opportunities in a commercially reasonable manner, and incorporate our conclusions into a valuation that quantifies the value we attribute to a target.

We aim to thoroughly assess the liabilities and costs, both operational and capital, associated with risks, as well as the potential cash flows linked to opportunities. If we determine that a risk cannot be effectively managed on a commercially reasonable basis, we refrain from pursuing the acquisition opportunity. Similarly, if the potential cash flow of an opportunity cannot be clearly determined, we do not include that cash flow in our valuation of a target. Our overall valuation for a target is derived from integrating our assessments of sustainability-related risks and opportunities, along with evaluations of other aspects of the target and its business.

Prioritizing Workplace Health, Safety & Well-Being

We believe that every person entering a Magris workplace should go home safe and healthy, every day. We also believe that every person at a Magris workplace is a courageous health, safety, and well-being leader who must act with a sense of urgency to manage our health, safety, including psychological safety, and well-being risks. All injuries and occupational illnesses are preventable, and there is no job worth doing in an unsafe way.

Workplace Health, Safety & Well-Being

How we define this topic: Managing occupational health and safety hazards involving physical and mental health; maintaining work environments that protect workers’ health and working capacity; and promoting workplace safety and health.

Our Approach

Safety is a core value at Magris. **“Everyone goes home, safe and healthy, every day”**—this applies to everyone who enters a Magris workplace. We believe that nothing is more important than the safety of people and that everyone has a role to play in creating a safe workplace.

We do not compromise on safety. We believe that all injuries and occupational illnesses are preventable, and there is no job worth doing in an unsafe way—none! While we strive for zero reportable safety incidents across our operations, we maintain a sharp focus on the complete avoidance of high-consequence incidents.

Our approach to managing safety is guided by three core concepts:



Leadership

Our leaders set the tone to demonstrate safety as a core value through visible leadership and clear communication of safety expectations.



Teamwork

We act as a team, and we expect everyone to be treated with respect, to trust and be trusted, and to listen to and support each other to work in a way that promotes well-being.



Practice

We act as a team, and we expect everyone to be treated with respect, to trust and be trusted, and to listen to and support each other to work in a way that promotes well-being.

Our Workplace Health, Safety & Well-Being Policy reflects our three core concepts and sets out the framework by which we protect everyone’s physical, psychological, and social health, safety, and well-being at our workplaces.

Reporting Safety Concerns

We encourage everyone to be safety champions and to address and report any unsafe practice or activity through our standard internal reporting channels (i.e., directly reporting to a manager). By reporting unsafe behaviours, everyone can learn from these experiences. We also maintain a third-party administered Code of Conduct reporting system so that anyone at a Magris workplace can anonymously report a safety-related concern or violation outside of our internal reporting channels.

We investigate all safety incidents thoroughly to understand the root cause and subsequently adapt our safety standards and practices to avoid any recurrence. We treat near misses* as safety incidents in our reporting and investigation process to help prevent future incidents that could be caused by similar circumstances.

Accountability

We incorporate safety-related objectives into our business objectives and tie individual compensation to the achievement of these objectives for many of our employees. As we progress towards utilizing leading indicators beyond our current lagging indicators, we continue to review how compensation can be best tied to safety performance. We also use safety-related criteria in our process to select our contractors.

* We base our definition of a “near miss” on the SASB definition: “an unplanned or uncontrolled event or chain of events that has not resulted in a recordable injury, illness, physical damage or environmental damage, but had the potential to do so in other circumstances”, except that we consider “other circumstances” to mean a slight shift in any one circumstance.

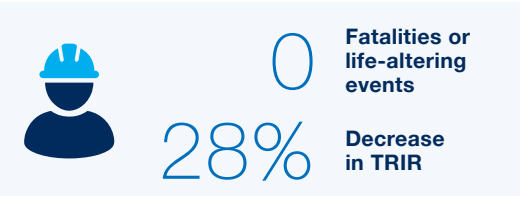
Triple Zero

We have a “Triple Zero” target – zero fatalities, zero lost time injuries and zero medical treatment injuries—that is a goal that we work towards every year. We track these using commonly accepted industry definitions.

Indicator	Unit	2022	2023
Total recordable injury rate (TRIR)*	Rate	3.97	2.85
Employees	Rate	4.23	1.79
Contractors	Rate	2.81	3.67
Near miss frequency rate (NMFR)†	Rate	1.98‡	2.06
Fatalities	#	0	0

* We define TRIR using the SASB definition: “the number of work-related injuries per 200,000 hours worked”.
† NMFR = number of near misses / (total hours worked / 200,000).
‡ 2022 NMFR has been restated (from 1.25) to reflect the subsequent harmonization of our tracking methodology across our businesses for this KPI.

Our 2023 Performance



Achieving our vision of “Everyone goes home, safe and healthy, every day” means that we are on a never-ending journey to continually assess the factors behind our Workplace Health, Safety & Well-Being performance, and act to improve it.

In 2023, we reported zero fatalities, alongside a total (employee and contractor) recordable injury rate (TRIR) of 2.85 and an overall near miss frequency rate (NMFR) of 2.06. Total and employee TRIRs decreased 28% and 58%, respectively--we believe that the improvement in both of our total and employee TRIR reflects in part our continued focus on and the impact of “visible felt leadership”--leaders being available, visible and felt in the workplace--in addressing high-severity risks across our operations during daily activities and in our health and safety training programs. With respect to the increase in contractor TRIR, the health and well-being of everyone at our workplaces is always our first priority. We will engage with our contractors in 2024 to address this aspect of our 2023 safety performance. Our 2023 total NMFR showed a slight increase over that for 2022; we believe the increase is within expected variances.

As part of our ongoing efforts towards “Triple Zero”, we focused on the management of high-consequence/life-altering incident risks. This approach is proven to be effective in increasing safety awareness and focus and fostering a strong workplace health, safety, and well-being culture. We also made progress towards improving our risk identification systems by standardizing critical controls for high-severity risks. We emphasized concrete actions to address both high-severity and low-consequence risks and believe this drove improvements in our overall year-over-year safety performance. Some key initiatives undertaken at our sites included:

At Niobec

- Increased focus on management and supervisory field presence, including reorganization of the health and safety team and the addition of a dedicated health and safety surface operations advisor;
- Conducted training focused on lifting and rigging, working at heights, and energy isolation;
- Revitalized lockout/tag out program; and
- Completed overhaul of crew pre-start meetings, with a focus on discussion of performance indicators and potentially fatal risks.

At Magris Talc

- Held “Safety Connect” days focused on “active caring”, which involves going above and beyond what is required or mandated to ensure the well-being and safety of others in the workplace. Active caring fosters a culture where individuals are attentive to the safety of their peers, proactively identify hazards and substandard conditions, and have the courage to approach their peers and intervene if they see someone putting themselves in danger because of an at-risk behaviour; and
- Increased focus on high-severity risks, including the use of monthly “Safety Blitzes”, where staff visit different operational areas and focus on conditions that could present high-severity risks, leading to greater discussion and awareness of risks.

We updated the Magris Workplace Health, Safety & Well-Being Policy during 2023 as part of our harmonization efforts. We rolled out the new policy in 2024, with the aim of harmonizing Niobec’s and Magris Talc’s operational standards, processes, and procedures to align with it.

We are also proud to note the recognition given by the Quebec Mining Association in 2023 to five Niobec supervisors for achieving in excess of 50,000 hours of operation without a reportable incident, one of whom achieved in excess of 100,000 hours without incident.



Our Future Actions

We will continue our work towards achieving “Triple Zero” during 2024 by continuing to improve our risk identification system and standardize critical controls for high-consequence risks. These efforts will include:

- Developing a critical controls pocket reference tool that addresses high-consequence risks at Niobec;
- Updating field-level risk assessment tools at Magris Talc;
- Continuing to standardize internal safety terminology and reporting indicators across our businesses;
- Creating a formal process to share lessons learned from relevant safety risks and incidents between our businesses; and
- Developing, defining, and documenting a set of leading indicators and associated tracking methodology and incorporating these indicators into our internal reporting.

In connection with our updated Workplace Health, Safety & Well-Being Policy, we will refine our associated management systems and processes in 2024 and subsequent years. These refinements will include, among other things:

- A management program that assesses and tracks prevention activity effectiveness over time;
- A system that monitors, tracks and assesses the effectiveness of safety training across Magris; and
- Timely and periodic evaluations of our health and safety management systems, processes and culture, including periodic external reviews.

Relevant Standards
SASB EM-MM-320a.1
GRI 403-8, 403-9, 403-10



Driving Safety Culture and Awareness Through Prestart Meetings

In workplace health and safety, proactive prevention activities can be highly impactful. Among such activities, an effective prestart meeting can be one of the most important moments of the workday. Gathering all team members before work commences to discuss the schedule for the day and ensure that everyone is ready to safely complete the day's activities can go a long way towards ensuring that “Everyone goes home, safe and healthy, every day”.

In 2023, Niobec conducted a review to improve and standardize the format of its daily prestart meetings. Largely led by supervisory personnel, this initiative resulted in the creation of an interactive prestart meeting board. Serving as an “information center,” it comprehensively covers all essential topics to be addressed before commencing the day's work. These topics include the previous 24 hours' health and safety, environmental, and operating performance as well as high-consequence risk identification, associated critical controls, and short-term corrective actions or plans pertinent to the day's activities. The revised format emphasizes raising awareness and discussion of performance indicators and potentially fatal risks.

The format was initially tested by a small group, and then progressively rolled out to all Niobec work teams. In the end, 137 employees received training on how to lead prestart meetings using the revamped format. The lead group of supervisory personnel continue to work on improving the meeting format and an action plan is being developed to ensure continuous improvement in this daily activity.

Business Continuity

How we define this topic: Understanding and responding to operational, environmental, social, cyber, and/or political events that could impact our business and our stakeholders. Minimizing the consequences of such events to allow for continued business operations.

Our Approach

Our business continuity efforts are critical to safeguarding people during and after any emergency or disruptive event that affects us and our ability to continue to produce and distribute our products. Our Business Continuity Policy sets out our framework to ensure that amidst an emergency or disruptive event, we can continue to supply materials to enhance performance and sustainability in critical products that improve everyday life in a lower carbon world.

It is crucial that we are prepared for any potential emergency or event that could cause disruption or endanger our people, facilities, or data. This readiness includes implementing measures to prevent such events from occurring in the first place. We take the following actions to put our business continuity plans into effect:

- Conduct simulated exercises to test our emergency preparedness, continuity, and recovery plans to ensure we can effectively respond to and address a real-life disruptive event;
- Revise our emergency preparedness, communications, continuity, and recovery plans following any material change in our operations, technology, processes, and/or personnel;
- Establish disaster recovery plans for applications and processes that support critical business operations;
- Establish crisis management teams that respond to a disruptive event and have nominated senior leaders with the authority to make critical decisions, which can include site closure or ceasing operations;
- Identify and mitigate supply chain risks to build resiliency and flexibility in our ability to source and procure the materials and services we need during and following a disruptive event;
- Maintain physical and technical safeguards to address identified information, data systems, and cybersecurity risks and conduct risk assessments as prudent, including external and internal penetration tests and social engineering campaigns;

- Train our employees regarding cybersecurity on a regular basis and as warranted by changes to our business operating environment or in response to identified risks; and
- Implement cybersecurity access procedures before we allow external parties, including service providers, to access our networks.

Our executive management team is responsible for non-operational business continuity risks while our operational management teams address operational business continuity risks.

Our 2023 Performance

In 2023, we updated our crisis management plans for each operation and MPM developed a corporate crisis management plan that is aligned with the existing plans in place at each of the operating businesses.

During 2023, our information technology (IT) groups worked to implement recommendations for remediating identified issues and improving the strength and security of our systems. These actions were informed by a comprehensive 360° review and audit of our cybersecurity and IT systems conducted in late 2022. As part of our cybersecurity risk management, we conducted multiple phishing simulations throughout the year, including within Niobec and Magris Talc, to test the awareness of our personnel and assess our overall risk of email-initiated fraud.

Our Future Actions

During 2024, we will conduct a simulation exercise to test our emergency preparedness and the operation of the corporate crisis management plan. We will also conduct another 360° review and audit of our cybersecurity and IT systems as part of our ongoing efforts to monitor and improve their readiness and performance.

We also plan to work on the following multi-year initiatives during 2024 and beyond:

- Complete a gap analysis of our crisis management plans against recognized standards; and
- Develop a crisis management preparedness standard addressing the mechanisms to:
 - Maintain up-to-date crisis management plans;
 - Provide training to ensure the preparedness of crisis management teams;
 - Test alert and information mechanisms on a regular basis; and
 - Conduct relevant simulations routinely.

Relevant Standards
SASB: EM-MM-530a.3





Niobec's Mine Rescue Team: Continuous Hands-On Learning

Niobec's mine rescue team is essential to its operations and its emergency response preparedness efforts. The team is made up of volunteer employees who annually commit to an intensive, specialized mine rescue training program that includes leading various operational missions and emergency simulations.

In 2023, the mine rescue team achieved all of the training objectives set by the Mine Rescue Service of the Quebec *Commission des normes, de l'équité, de la santé et de la sécurité au travail* (CNESST), the provincial regulatory authority for occupational health and safety. The mine rescue team also received a new, specialized mine rescue vehicle that will enable the team to conduct faster and more efficient interventions. The size of the team was also expanded from 24 to 36 individuals, demonstrating our employees' commitment to emergency preparedness.

From the larger group, Niobec selected a small team of elite rescuers to participate in the annual provincial mine rescue competition organized by Mine Rescue Service. The qualification process for this competition entails a rigorous training program and a preliminary competition to identify six finalists. In 2023, Niobec's elite team was one of the six finalists selected for the final competition.

Participating in the final competition provided the team with a valuable opportunity to share best practices with other mine rescue teams from across the province. It also reflected the team's effort and dedication to achieving the highest levels of emergency response preparedness.

Respecting the Environment

The preservation of the environment is fundamental to our present and future success. We are committed to avoiding, minimizing, or offsetting, in that order, all environmental impacts associated with our operations, including the greenhouse gases we generate or can influence.

GHG Emissions & Climate Change

How we define this topic: Supporting the transition to a lower-carbon economy by managing and monitoring our direct and indirect contributions to climate change (including GHG emissions), understanding and addressing climate-related risks and opportunities, and employing adaptation and resilience measures.

Our Approach

Climate change is a challenge to human health, well-being, and our planet, and we believe everyone has a responsibility to respond. As a business, we have a critical role to play in:

- 1. **Taking action** to reduce and offset greenhouse gas (GHG) emissions that are within our control;
- 2. **Engaging** with others on GHG emissions mitigation and adaptation activities;
- 3. **Working** to make our businesses resilient to the effects of climate change; and
- 4. **Producing** products that enable and result in low-carbon solutions.

Our risk management process accounts for climate-related risks and opportunities, while the identification and mitigation of these risks are integrated into our business continuity, strategic planning, and capital allocation processes. In recent years, we have increasingly engaged directly with our value chain partners to better understand how we can minimize our upstream and downstream GHG emissions, and plan to expand these efforts. We also work directly with our customers to ensure that our final products are optimally designed for use in their own products.

Although we are not a major emitter, we believe that it is still important to reduce our GHG emissions. In addition to tracking our GHG emissions, we have programs that support reduction by increasing energy efficiency and evaluating the use of equipment that does

not rely on fossil fuels. Our GHG Emissions & Climate Change Policy sets out our framework to improve the efficiency and sustainability of our operations and products.

Magris is proud to support its customers to enable them to produce lower-carbon alternatives, as indicated by our Niobium LCA Study. The study concluded that the carbon footprint for our FeNb production is 15.96kg CO₂e/kg. It also concluded that the GHG emissions avoided by the niobium we supply to our customers are 58 times the amount of the GHG emissions we generate from producing it. For more information, please refer to the “[How We Improve Everyday Life](#)” section of this Report.

Our 2023 Performance



Over 85% percent of the power that our facilities purchase is generated from renewable sources. In 2023, our Scope 1 and 2 emissions were 47,753 t CO₂e and 11,644 t CO₂e, respectively. We saw a 2.8% increase in Direct GHG emissions driven by an increase in Magris Talc’s direct emissions due to increased waste stripping at the Yellowstone mine, partially offset by a slight decline in Niobec’s direct emissions as a result of warmer winter weather. These factors also contributed to the increase in Talc GHG intensity and the decrease in Niobium GHG intensity.

Magris GHG Emissions (Scope 1 and Scope 2)	Unit	2022	2023
Direct GHG emissions (Scope 1)	tCO ₂ e	46,474	47,753
Indirect GHG emissions (Scope 2)	tCO ₂ e	12,376	11,644
Talc GHG intensity (Scope 1 and Scope 2)	tCO ₂ e / t	0.13	0.16
Niobium GHG intensity (Scope 1 and Scope 2)	tCO ₂ e / t	1.61	1.31

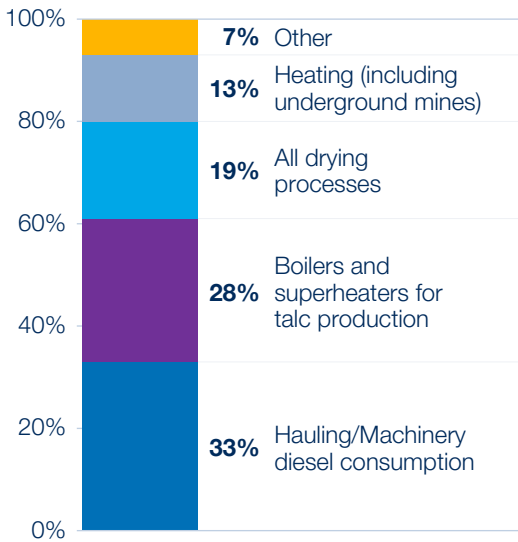
We are committed to achieving net-zero emissions by 2050 at the latest. We recognize that this is a serious undertaking and continue our work to fully understand all of the actions we will need to take to achieve this goal. In 2023, we made progress against our interim objectives including establishing a standardized tracking methodology for Scope 1 and Scope 2 emissions. This involved documenting the methodology used for our data collection protocols to ensure consistency across Magris. For Scope 1 emissions, we track our fossil fuel consumption and calculate emissions based on total consumption for each type of fuel using the latest relevant emissions factors. Scope 2 emissions are calculated using a market-based approach where data is available and a location-based approach where it is not or where we believe the data may not be accurate, utilizing the most recent emissions factors available.

One of our interim goals is to implement a comprehensive Scope 3 emissions framework. Due to the complexity of our products’ contribution to emissions, our focus is prioritizing an understanding of where in the value chain we can influence emission reductions.

To advance this, in 2023, we developed a framework to assess our value chain, focusing on understanding our upstream and downstream emissions.

As part of these efforts to establish our roadmap for emissions reductions, we identified and documented, the main 2023 GHG contributors at all our operations.

Primary Contributors to Scope 1 Emissions



Our Future Actions

Our path to net-zero includes continuing work to establish a net-zero carbon emissions roadmap by the end of 2024. This involves:

- Creating an internal Climate Change Taskforce with a mandate to develop the net-zero carbon emissions roadmap, including interim targets by the end of 2024. Additionally, the Taskforce will identify tangible actions and opportunities for GHG emission reductions and assess the associated risks and financial impact. Furthermore, it will develop an action plan for short-, medium-, and long-term emission reduction objectives; and
- Further developing an emissions framework that will enable us to fully engage with our value chain partners to better understand our Scope 3 emissions and assess where in the value chain we may have the ability to influence or drive emissions reductions.

Relevant Standards
SASB: EM-MM-110a.1, EM-MM-1101.2
GRI: 305-1, 305-2, 305-3, 305-4, 305-5



Magris Talc Boosts Energy Efficiency With New Fluid Bed Dryer

The Magris Talc Three Forks Milling operation commissioned a new fluid bed dryer for its pelleting process in April 2023, enabling greater energy efficiency in our operations. The new fluid bed dryer uses a simplified integrated dryer/cooler system with a single burner and two fans, replacing an older design system with five fans and three burners and improving the drying capacity and efficiency of the fluid bed dryers and cooler.

The new system features a variable frequency drive, enhancing energy efficiency and permitting precise control of dryer air flow. This upgrade has resulted in a notable reduction in operational load, from approximately 150 kW to 130 kW, representing a 13% reduction in electricity usage. Based on increased throughput achieved with the new system, we also expect to increase natural gas consumption efficiency by 8% to 15%, depending on the product being made.

Water & Effluent Management

How we define this topic: Managing and monitoring our use and conservation of water resources to meet our needs while providing accessibility to others, including the availability of freshwater, reduced water consumption, and the treatment and re-use of wastewater and greywater.

Our Approach

Based on our core value of Respect, we understand the need to conserve and protect all natural resources that we use or that are impacted by our activities. Water is a shared resource that we rely on throughout our operations, particularly at Niobec, and that is also vital to the well-being of the environment and the communities adjacent to us.

Our water and effluent management practices are highly regulated, particularly concerning effluent quality and quantity. We maintain transparent, honest, and respectful communications with regulatory authorities and engage proactively with them regarding any situation that could result in a potential non-compliance. We work to ensure that our actions and the solutions we implement meet their expectations and all regulatory requirements. Our Vice President, Operations maintains oversight of our Water & Effluent Management Policy, which sets out our framework to reduce our footprint in the watersheds where we operate through responsible use and stewardship of water.



Efficient and responsible use, treatment, and discharge of water and effluents is critical to maintaining our operational sustainability. Since water is a shared resource, we continuously work to reduce our footprint in the watersheds where we operate. Each of our operations employs a water management strategy and works with regulatory authorities and water users of the local watershed. Our work to reduce our watershed footprint involves four aspects:



Water Quality

We invest in infrastructure to improve the quality of our effluent management processes and ensure continuous compliance with applicable laws and regulations. Our compliance is confirmed through regular mandated testing, monitoring and reporting under various regulatory regimes. Water quality is also a key consideration in our operational and capital expenditure decision-making processes.



Water-related Interactions

We proactively raise water-related issues and concerns with regulators in order to collaborate on appropriate responses and solutions. In addition, we identify people and communities who may potentially be affected by our water use and management. We communicate with these stakeholders as issues arise.



Water Quantity

We prioritize efficient water use and water conservation to minimize the amount of intake water we need to meet our operational needs. To do this, we set and revise short-term and long-term targets regarding intake water quantities, conservation and recycling of water, and effluent quantities, using a context based approach that recognizes individual watersheds as system boundaries. We also incorporate infrastructure to manage the quantity of our water and effluent outflows to reduce these outflows and comply with regulatory requirements.



Climate Change Resiliency

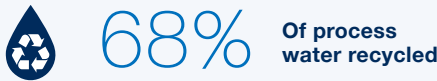
We look to build resiliency to our operational processes and infrastructure to address the real and potential impacts of climate change on our watersheds from greater weather variability and extreme weather events.

Addressing these four aspects requires us to integrate the financial, environmental, and social values of water into our operational decisions for project development, business planning, and mine closure in addition to proactively assessing ongoing water risks and opportunities.

While we aim to standardize our water management practices throughout our operations, we continue to expect higher engagement from operations where this topic is of higher importance. One such case is Niobec, whose integrated water and effluent management approach is guided by industry best practices as well as the Water Stewardship Protocol from the Mining Association of Canada TSM Initiative. Meanwhile at our Yellowstone and Vermont facilities, removal of non-economic material from ore (beneficiation) is by dry separation which means these facilities use no water in their processes.

To monitor, understand, and refine our water and effluent management practices, we set KPIs that are relevant to the context of each operation. We use these KPIs to track our performance and progress in achieving our targets. Additionally, we regularly report on our water performance internally and share our monitoring results as mandated by relevant regulations.

Our 2023 Performance



One of our key commitments is to avoid any water-related notices of violation (NoVs). 2023 was the first full year of operation of Niobec’s integrated water management facilities and approach. These facilities include a \$33 million water treatment plant, a leachate neutralization circuit that neutralizes hydrochloric acid, and a paste-backfill plant that enables chloride-containing brine to be safely sequestered underground.

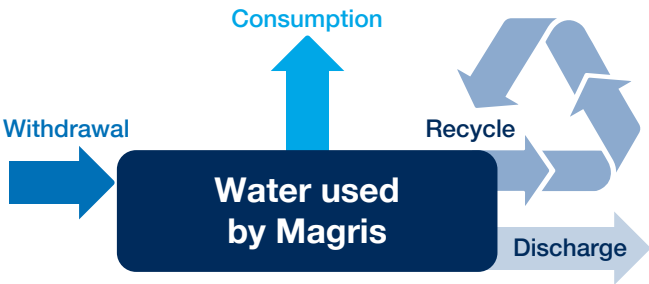
Our Penhorwood talc mine received one NoV from the Ontario Ministry of the Environment, Conservation and Parks (ECP) in 2023. Although Penhorwood treats its effluent prior to discharge, concentrations increased during transit through historical tailings, resulting in exceedance of permitted limits. Efforts were made during the year to address this challenge; however, effluent concentrations at certain points in 2023 exceeded the limits allowed under Penhorwood’s permits. Penhorwood is actively working with provincial and federal regulatory authorities, including the ECP, Environment and Climate Change Canada, and Department of Fisheries and Oceans, to achieve full compliance. A critical part of Penhorwood’s effort has been engagement with the local Indigenous communities to understand their concerns and evaluate the potential impact on the communities of any solution put forward.

At Niobec, completing the necessary studies and obtaining regulatory authorization for a permanent solution to manage pluvial effluent (run-off) took longer than we expected; however, we anticipate that a permanent pumping installation will be commissioned and operational by the end of 2024. We continued to use an interim pumping installation to pump pluvial effluent within our existing water management system while we worked on the permanent installation.

While working to harmonize internal water management reporting across Magris, a task we hoped to complete in 2023, we concluded that complete harmonization would not yield greater benefits than maintaining the separate reporting regimes for Niobec and Magris Talc. As a result, we shifted our focus to harmonizing specific elements of the reporting, where appropriate, a task we will continue in 2024.

Throughout 2023, we also worked to identify opportunities to optimize water use and recycling across Magris. This initiative will continue into 2024 and we remain focused on doing so without compromising our water-related compliance. We aim to implement the identified opportunities while upholding this standard in 2025.

Indicator	Unit	2022	2023
Total water withdrawn from source bodies	megaliters	7,635	7,761
Total water used including recycled water	megaliters	25,184	24,321
Percentage of withdrawn water that is recycled	%	70	68
Total water discharge including treated water	megaliters	8,709	8,466
Water-related notices of violation	count	1	1



Our Future Actions

Full compliance with our regulatory obligations and avoidance of any water-related NoV is a core objective for us. In 2024, this will involve two key initiatives:

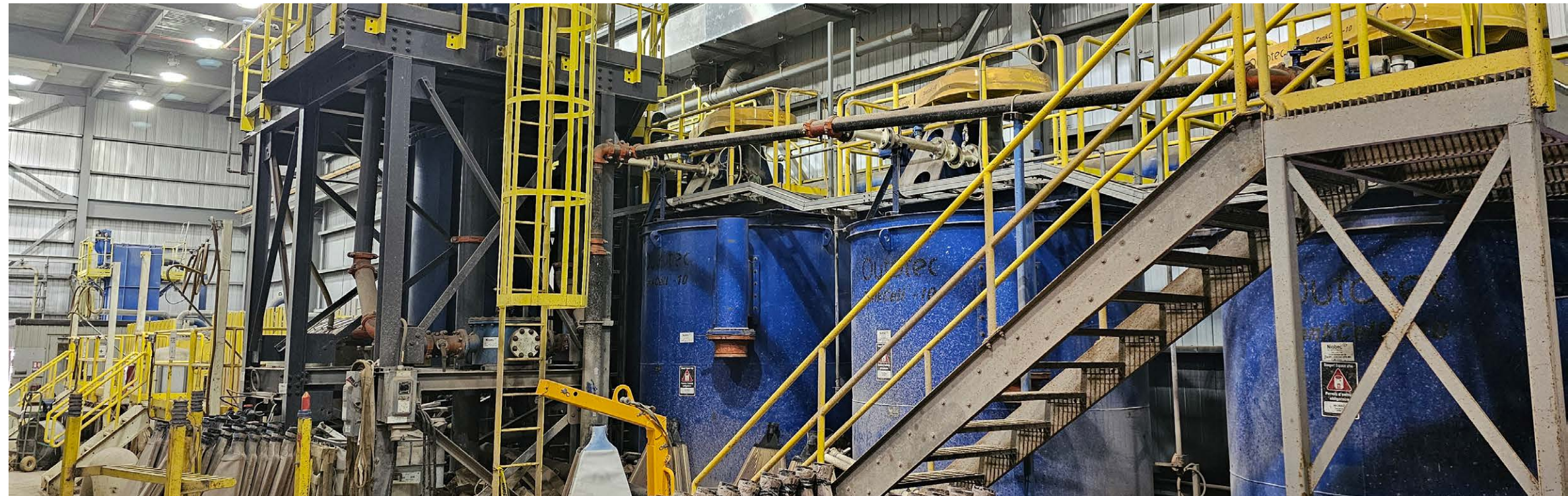
- Fully operating a permanent pluvial effluent pumping station at Niobec by year-end; and
- Achieving a solution that addresses the challenge to manage tailings effluent to discharge standards at Penhorwood. Penhorwood will continue to engage with the local Indigenous communities regarding their concerns and work to complete various milestones agreed upon with the ECP, including a voluntary action plan, while keeping the federal regulatory authorities informed.

Although we concluded that complete harmonization of internal water management reporting would not lead to significantly greater benefits, we plan to complete the beneficial aspects of harmonization that we identify and set operational targets for our internally reported water management indicators.

More information on our water and effluent management can be found in the “[More About Magris](#)” Appendix of this Report.

Relevant Standards

SASB EM-MM-140a.1, EM-MM-140a.2
GRI 303-3, 303-4, 303-5, MM3



Niobec Paste Backfill Plant

In underground mining, paste created from mine tailings is a method that is often used to fill in mined-out stopes to provide structural support and safely optimize mining in areas adjacent to previously mined areas. This can also benefit tailings management since tailings used in paste are diverted away from surface tailings ponds or facilities, thereby reducing the capacity requirements for surface storage. Niobec annually diverts approximately 40% of its tailings (approximately 920,000 t) away from surface storage through its use of paste backfill.

Uncommonly, Niobec’s use of paste backfill is also an important part of its water and effluent management. With improvements made to the paste backfill plant as part of Niobec’s integrated chloride management, the plant can use the chloride-containing brines and HCl leachate produced by Niobec’s water treatment processes to make paste. This eliminates the need to further treat the brines and leachate.

Using brine and leachate rather than process or other water complicates the paste production process, and required repeated experimentation and testing to achieve an optimal paste recipe that maximized the amount of brine and leachate used without affecting the consistency and other necessary properties of the paste.

Once the paste is pumped into an empty stope, it quickly begins to harden and cure, which allows the brines and leachate to remain safely stored within the hardened paste without risk of future release into underground water.

For all of these reasons, we consider Niobec’s use of paste backfill produced from brines and leachate to be a prime example of how Magris operates responsibly.

Waste Management & Product Stewardship

How we define this topic: Handling, storage, transport, disposal, and diversion of non-hazardous and hazardous waste, including tailings waste. Managing and monitoring the non-renewable and renewable materials we use to manufacture and package our products, including the measures we take to retain the value of such materials for as long as possible.

Our Approach

Our core value of Respect, including for the natural world and those affected by our activities, underlies our commitment to minimize any adverse waste impacts that our activities and products may have. We aim to reduce or eliminate our waste streams where possible to retain the value and extend the life of the materials we use. Where necessary, we manage our waste streams through proper handling, storage, transport, disposal, and diversion. Our Waste Management & Product Stewardship Policy establishes our framework for the management of our waste and for our product stewardship efforts to enable us to achieve our waste management and product stewardship goals.

Our efforts to reduce our environmental impact through waste management are interconnected with our product stewardship pursuits. These practices enhance the sustainability of our business through more efficient use of materials and resources, which in turn supports our business resilience and continuity. In most cases, our customers integrate our products into their own products in small amounts. As a result, our products are inseparable from their own. For this reason, our waste management and product stewardship efforts focus on implementing measures to reduce the adverse impact and use of materials and resources we use within our own processes. We implement a circular economy approach to emphasize keeping materials, products, and services in circulation for as long as possible.

Because of the diversity of our waste streams, we require a variety of methods to manage them properly. We classify our streams of waste into three categories: non-hazardous waste, hazardous waste, and mineral waste (waste rock, tailings, and overburden).

Non-hazardous Wastes

We manage our non-hazardous waste using the reducing, reusing, recycling, valorizing, and disposing hierarchy. While the concept is well known, applying it to our industrial operations requires continuous engagement with our workforce and with our supply chain partners. Based on our core value of Excellence, we aim to reduce the amount of non-hazardous waste that we send to landfill by finding ways to divert waste (mainly packaging) from landfill.

Hazardous Waste

Our approach to hazardous waste management is based on industry best practices. We regularly evaluate the environmental and workplace hygiene risks of any hazardous products before using them in any of our processes. Our internal hazardous waste management standards and practices are based on applicable laws and regulation and industry best practices. Where possible, we prioritize using non-hazardous substitutes for hazardous materials in our operations. We also use service providers that possess all relevant certifications for responsible disposal of hazardous wastes.

Tailings Wastes

We are committed to using responsible mine waste management practices and systems to safely store and manage our tailings, to prevent any unexpected event and to minimize risks to the environment and society even though the geochemical characteristics of our tailings do not pose significant risks to the environment due to the nature of the minerals we mine. Our Tailings Management Policy describes our approach to managing tailings and outlines our requirements for annual inspections and reviews of our tailings facilities risks; internal and external reporting of our tailings management performance; and allocation of oversight responsibilities between the board, executive, and operational levels.

We manage our tailings waste using the principles, management framework, guidance, and tools contained in the [Mining Association of Canada’s Guide to the Management of Tailings Facilities](#). We also follow the [Global Industry Standard on Tailings Management \(GISTM\)](#), the Dam Safety Guidelines, and related technical bulletins of the Canadian Dam Association.

At all of our mines, we typically stockpile overburden topsoil for use in reclamation at the end of a mine’s life. At Magris Talc, waste overburden and waste rock are handled in one of three ways: crushed and used for capping and stemming; used for dam building if it has the necessary attributes; or placed in a permitted dump if it cannot be used for another purpose. There is generally no waste rock produced at Niobec and when there is, it is left in the underground mine.

Our Performance

Indicator	Unit	2022	2023
Total non-mineral waste produced	t	5,235	3,780
Total non-hazardous waste	t	5,018	3,495
Diverted from landfill	%	43.5	46.8
Total hazardous waste generated	t	217	285
Recycled	%	19.5	8.4
Valorized	%	49.3	40.7
Total diverted from landfill	%	68.8	49.1
Total mineral waste produced	t	5,802,013	6,259,074
Tailings	t	2,726,772	2,583,084
<i>Used for other purposes, including paste-backfill</i>	%	56.3	57.5
Waste rock and overburden	t	3,075,241	3,675,990

In 2023 our total waste generated was 6,262,854 t and primarily composed of tailings and waste rock overburden. We were able to divert 46.8% of our non-hazardous waste, recycle or valorize 49.1% of our hazardous waste, and repurpose 57.5% of our tailings. We consider the year-over-year differences for the hazardous and non-hazardous waste amounts between 2023 and 2022 to be within the variances we expect as they are largely influenced by the nature of our maintenance and operational activities and the nature of the waste generated. The increase in waste rock in 2023 resulted from an increase in overburden at the Yellowstone mine, which was partly offset by lower overburden at the Argonaut mine. We expect this trend to continue in 2024.

We produced a lower amount of tailings at Niobec in 2023 due to reduced throughput and higher ore grades. We expect to continue to increase production at the Niobec paste backfill plant over the next few years, which should allow us to send more tailings for storage in the underground mine instead of to the tailings facility.

Throughout 2023, we worked to standardize our waste reporting methodology across our operations with the goal of improving reporting on these waste streams and making more informed management decisions. While we hoped to complete this exercise by the end of 2023, some work remains, principally reconciling differences in waste categorization, tracking, and management due to local regulations and locally available waste management services.

We conducted gap analyses against the GISTM standard at each of Niobec and Magris Talc. The analyses helped us better understand and document the efforts and investments required to meet the standard. This is especially relevant to Penhorwood where we constructed an enhanced tailings facility (in accordance with industry best practice) in conjunction with a dam lift.

We received the final regulatory inspection for the first lift of the Penhorwood tailings dam. This was an essential step in upgrading the functionality of the tailings facility at the mine.

We identified opportunities across Magris for greater recovery, reuse, and recycling of production and packaging materials. This work has resulted in a further effort to study and/or implement various initiatives on an operation-by-operation basis aimed at reducing our waste production, including the volcanic ash initiative at our Yellowstone mine discussed under “Our Future Actions” below.

Niobec renewed its waste management certification under the Quebec provincial performance recognition program “ICI On Recycle +”, receiving the highest level of certification (Elite).

Our Future Actions

In 2024, we will continue with our efforts to reduce our waste streams. These activities will include:

- Assessing the tracking of waste streams to fully capture, understand, report on, and improve our waste management performance at Magris Talc, to better differentiate between various waste streams. We expect this undertaking to take several years to complete;
- Revising our tailings management processes and mechanisms to implement the findings of our GISTM gap analyses where we believe we can achieve a meaningful improvement; and
- Continuing to identify and assess potential opportunities across Magris for greater recovery, reuse, and recycling of our resources and our production and packaging materials. For example, the Yellowstone Mine is working with local cement producers to evaluate substituting its volcanic ash produced as waste for fly ash in the cement production process. This evaluation is expected to be ongoing into 2025. If successful it would allow Yellowstone to divert a significant amount of volcanic ash from its waste stream.

Relevant Standards
SASB EM-MM-150a.2-10, EM-MM -540a.1-2
GRI 301-1, 301-2, 301-3, 306-2, 306-3, 306-4, 306-5



Giving a Second Life to Niobec Waste

As an industrial operation with high operating standards, Niobec generates materials that we view as waste but that could be of use to others. As part of Niobec’s strong recycling culture, Niobec continues to look for opportunities to give a “second life” to the waste materials it generates. An example of Niobec’s efforts in this area is around used fan pipes. Despite facing challenges in recycling the used fan pipes, which contain fire retardant plastic, Niobec remained committed to finding innovative solutions and stored the fan pipes in anticipation of eventually finding an opportunity to recycle or reuse them.

In 2023, after four years of storage, Niobec collaborated with a plastic recycling facility to transform the fan pipes into smaller pipes that can be used to load explosives in its mining operations.

This endeavour not only prevented the disposal of fan pipes into landfills, but also minimized the need for new materials for explosives loading. Given the widespread use of similar fan pipes across the mining industry, we believe that this initiative has potential for broader application and can set a precedent for sustainable practices in the mining industry.

Respecting People

Magris’ core values of Respect, Integrity, Safety, and Excellence are rooted in the relationships we strive to build and maintain with people—our employees, contractors, supply chain partners, customers, communities, and others. We must treat people with honesty and respect, think through how our actions or decisions can affect people, and act appropriately. We are committed to fostering and contributing to the communities in which we operate through transparency, consultation, and collaboration.

Workplace Culture

How we define this topic: Supporting diversity, equal opportunity and inclusiveness in our businesses; promoting a collaborative, supportive, and respectful work environment to increase participation and collaboration amongst our people; appropriately reflecting our communities; advocating for equal pay and opportunities; and engaging in anti-discrimination efforts.

Our Approach

Respect is one of our four core values and encompasses respect for each other, the natural world around us, and all those with whom we interact or that are affected by our activities. It is what leads us to seek a diverse, equitable, and inclusive work culture where every person is welcome, respected, and treated with dignity regardless of our differences in age, ancestry, colour, race, ethnicity, ethnic origin, place of origin, creed, disability, family status, marital status, gender identity or expression, sex, or sexual orientation.

We understand that fostering a culture that promotes a diverse, equitable, and inclusive work environment makes our business stronger and more sustainable.

We also believe that our ability to innovate and make good decisions is enhanced by having diverse perspectives, and that our workforce is more engaged and collaborative when we reflect the diversity of our communities, treat people fairly, and include everyone.

Our Workplace Policy establishes the framework through which we promote respectful, fair, and inclusive workplaces based on Respect, Integrity, Safety, and Excellence. This involves:

Diversity

- Drawing from a broad talent pool when possible within the local region to reach outstanding talent and have a workforce that reflects the communities where we operate.

Equality

- Employment-related decisions that are free from discrimination;
- Addressing potential unconscious bias in other employment decisions and talent practices (including performance and development, compensation, hiring); and
- Zero tolerance of discrimination—different or less favourable treatment of any individual or group based on the personal characteristics described above—within our workplaces and in all aspects of the employment relationship with us.

Inclusion

- Zero tolerance of violence, bullying, or harassment in our workplaces and in all aspects of the employment relationship with us;
- Reasonable accommodation for individuals with a disability, those with needs related to their religious observance or practices in recognition of personal religious expression, and of other personal characteristics as required by applicable laws; and
- Supporting behaviours that acknowledge and respect our differences, value the different perspectives and points of view that come from those differences, and encourage listening and considering those different perspectives and points of view in our work.

Accountability

We expect our leaders to take an active role in fostering an equitable, diverse, and inclusive culture in our workplaces. This means consistently displaying inclusive leadership behaviours, including valuing all perspectives, modeling inclusive and respectful conduct in the work environment, and encouraging collaboration.

We also encourage everyone to report behaviours that are not consistent with a diverse, equitable, and inclusive workplace through our various reporting channels. We take all reports seriously and investigate them thoroughly, and if a violation of our Code of Conduct or values has occurred, we take appropriate disciplinary action.

Recruitment

Our recruitment practices and decisions reflect our core value of Respect and are guided by the principles set out in our Workplace Policy.

Our 2023 Performance

Indicator	Unit	2022	2023
Total employees	#	848	856
Total female employees	#	125	124
Total female employees	%	15	14
Total Leadership level employees	#	160	178
Leadership level female employees	#	36*	39
Leadership level female employees	%	23	22
Employees under 30	% of total	14	12
Employees 30-50	% of total	57	55
Employees 50+	% of total	29	33

In 2023 we had 856 full time employees, of which 124 or 14% were women. Twenty-two percent of our leaders† were women.

In 2023, we conducted and completed diversity, equity, and inclusion training for all employees with managerial responsibilities, and achieved a 100% completion rate for this training. To ensure we are aligning our actions with our beliefs, we also began a process to identify and remove any exclusionary language used across Magris. As a first step in the process, all titles at Niobec were revised to remove the use of exclusionary language.

* Restated (from 45) to reflect correction of a data error.
† We define a leader as someone at the supervisor level or above.

Our Future Actions

We intend to take the following actions to continue to promote an inclusive and respectful environment at our workplaces:

- Conduct and complete in-depth diversity, equity, and inclusion training for all employees in 2024; and
- Continue with our process to remove all exclusionary language across Magris. We expect that this work will extend beyond 2024. As a next step, this process will involve removal of exclusionary language in all signage used at our workplaces and raising awareness of the negative consequences of using exclusionary language. We will then revise all of our written communications and materials (employee information materials, training and operating manuals, operating standards, etc.)

Relevant Standards
GRI: 405-1; 405-2



Business Ethics & Transparency

How we define this topic: Ensuring transparency in all tax, financial and record-keeping practices; compliance with laws and regulation, including those relating to bribery, corruption, anti-competitive behaviour and trade controls (sanctions); and a system that encourages the reporting of non-compliances and implements corrective actions to prevent recurrences.

Our Approach

Integrity is one of our four core values and is at the heart of ethical behaviour. This principle, along with our other core values of **Respect**, **Safety**, and **Excellence**, is what enables us to create trust and credibility amongst ourselves and with others to maintain strong, enduring relationships that are key to a successful and sustainable business.

We expect everyone at Magris to think through how their actions affect others, to act appropriately, and to hold themselves and others accountable.

Our policies, practices, and standards consistently reinforce the need to act ethically in all aspects of our business. The principal overarching tool that we use to guide ethical behaviour standards at Magris is our Code of Conduct, the full text of which can be accessed at <https://www.magrispm.com/sustainability>.

Our Code of Conduct serves as a resource to help everyone understand how they should conduct themselves while carrying out their duties at Magris. In addition, the Code of Conduct points to different help channels for clarification, decision-making, or reporting instances of behaviour that may be unethical or in violation of the Code of Conduct. It extends to everyone at Magris, including senior management and board directors. We also expect our contractors, supply chain partners, consultants, agents, and others who act on our behalf to recognize and abide by our Code of Conduct.

We do not operate or produce in any countries that have been identified by the Transparency International [Corruption Perceptions Index](#) as having high levels of perceived public sector corruption.

Our 2023 Performance

We received two reports of alleged violations of our Code of Conduct in 2023. Both reports were fully investigated and resolved without any substantiated misconduct. We experienced no incidents of corruption, bribery, or other violations of the Code of Conduct concerning legal compliance, including anti-trust violations and anti-competitive behaviour in 2023. We continually encourage everyone to report any concern or behaviour they believe is unethical using our standard reporting channels or our third-party administered Code of Conduct reporting system.

Ninety-five percent of Magris Talc employees and 75% of Niobec employees received training on our Code of Conduct and the Code of Conduct reporting system. This fell short of our objective for all employees to receive the training. As of the date of this Report, we continue with our efforts to have all employees, including new hires, receive this training. This training was carried out through our third-party service provider, ClearView Connects, which administers the Code of Conduct reporting system. In addition, over 95% of employees with responsibilities that touch on supply chain, sanctions, anti-trust, and anti-corruption concerns completed internally-led training for those topics.

In 2022, we voluntarily joined the UNGC as a demonstration of our commitment to sustainability in our businesses and will file our inaugural Communication on Progress in 2024.

Magris recently filed its inaugural report under Canada’s Fighting Against Forced Labour and Child Labour in Supply Chains Act. The Act requires businesses to report annually on the steps they take to prevent and reduce the risk of forced labour or child labour within their operations and their supply chains. You can reference Magris’ report at <https://www.magrispm.com/sustainability>.

Magris also reports annually under Canada’s Extractive Sector Transparency Measures Act (ESTMA), which is intended to deter corruption by increasing transparency around payments made to governments. ESTMA requires disclosure of certain reportable payments over C\$100,000. For a copy of our most recent ESTMA report, please visit our website at <https://www.magrispm.com/sustainability>. In aggregate, Magris’ payments to governments totalled approximately C\$7.29 million in 2023, compared to C\$9.1 million in 2022.

Our Future Actions

In 2024, we plan to:

- Review our Code of Conduct to determine if an updated version is required;
- Require all employees to acknowledge in writing their obligations under our Code of Conduct; and
- Refresh our Code of Conduct-related training as needed.

Relevant Standards
SASB: EM-MM-510a.1, EM-MM-510a.2 GRI:
205-1, 205-2, 205-3, 206-1, 207-2, 207-3, 207-4

Responsible Supply Chain

How we define this topic: Working to prevent and mitigate negative social and environmental impacts and create mutual benefit in our supply chain.

Our Approach

Supply chains are of critical importance and can present numerous risks and opportunities to any business. As a producer and supplier in global markets, it is imperative that we manage the legal, social, ethical, and environmental elements of our supply chain. Doing so demonstrates that we are acting consistently with our values to maintain and strengthen the sustainability of our business.

Our responsible supply chain framework integrates our key sustainability commitments into the overall management of our supply chain. We prioritize working with supply chain partners who share our values and commitment to operating responsibly and ethically. We also focus on potential impacts related to human rights, health and safety, the environment, and legal and regulatory compliance in the areas of anti-corruption, anti-competition, and sanctions as critical risks in our supply chain. Our Responsible Supply Chain Policy establishes our framework to ensure we have a responsible supply chain. The policy integrates our key sustainability commitments into the overall management of our supply chain.

Supply Partner Relationships

Our ability to operate requires a strong and dependable network of supply chain partners. To prevent and mitigate supply chain risks that could adversely impact people, the natural world, or our reputation, we set and use evaluation criteria aligned with our values and sustainability commitments to choose our business partners.

We work with our partners to identify, assess, and mitigate potential risks in our supply chain. This includes prioritizing local procurement of materials and services and communicating and reinforcing our expectations via our Supply Chain Partner Expectations Statement. To uphold our values, we are prepared to decline or withdraw business with any potential or current supply chain partner who does not meet our expectations or comply with legal and/or regulatory requirements.

We have also started a process to periodically assess our supply chain processes and partners to ensure compliance with our requirements. Furthermore, we are educating and training our procurement, logistics, and related staff in responsible supply chain practices.

Our 2023 Performance



43%

Of procurement budget spent locally

Local suppliers are a key part of our business. In 2023, 43% of our procurement budget was spent on local suppliers*, compared to 36% in 2022.

During 2023, as part of our evolving efforts to ensure our values are shared and implemented by our partners throughout our value chain, our Supply Chain Partner Expectation Statement was sent to all of our suppliers at Magris Talc and all of our logistics providers at Niobec. Additionally, select customers for both Magris Talc and Niobec received the statement. We requested acknowledgment and adherence to the Supply Chain Partner Expectation Statement from each of our value chain partners.

* Niobec defines a local supplier as one based within 100 kilometres of the operation. Magris Talc defines a local supplier as one located within the same state or province as the operation.

Overall, we had strong engagement from our customers and suppliers. More than 85% of Magris Talc’s suppliers and 75% of Niobec’s logistics providers signed and returned our Supply Chain Partner Expectation Statement. For customer responses, approximately half of Niobec and Magris Talc customers participated, with around 40% signing and returning our Supply Chain Partner Expectation Statement. Approximately 10% responded explaining that their own Code of Conduct or policies aligned with the principles in our Supply Chain Partner Expectation Statement. For those value chain partners who did not respond to our request, we will continue our efforts to engage with them during 2024. Our ongoing objective is to achieve 100% of our value chain partners in alignment with the principles set forth in our Supply Chain Partner Expectation Statement.

As a result of personnel and other administrative changes within Niobec’s procurement group during 2023, our Supply Chain Partner Expectation Statement was not sent to Niobec’s suppliers during the year. With a renewed team and focus on this initiative, we have developed an action plan that will be implemented during 2024 to further engage with Niobec’s suppliers on this topic.

We also started work to establish a formal framework for supply chain partner compliance reviews and audits. As a first step, we created a screening tool that we intend to use across our value chain to identify higher-risk counterparties (both suppliers and customers). When the screening tool signals a potential “red flag” associated with a counterparty, we will focus on and conduct a more detailed investigation into any potential risks to understand their implications. We will then determine what further action, if any, we should take. We expect that this process will evolve into a framework enabling us to conduct periodic reviews and, where required, compliance audits of our value chain partners. Given the large number of value chain partners we have for an organization of our size, our initial efforts in using the screening tool will be guided by several factors. These include the level of business we conduct with our counterparty, their location, the nature of the supply to us, and how our product is used by the counterparty.

Our Future Actions

We intend to take the following actions by the end of 2024 to further prevent and mitigate negative social and environmental impacts and create mutual benefit in our supply chain:

- Utilize our newly implemented screening tool to screen the top 20% of suppliers and customers at Niobec, and the top 10% of same at Magris Talc, based on dollar sales volume;
- Identify counterparties of concern and carry out detailed diligence reviews of them where indicated by the presence of screening tool “red flags”; and
- Require all new suppliers to provide a written acknowledgement of our Supply Chain Partner Expectations Statement before entering into business with them.

Relevant Standards
GRI: 204-1

Content Indices

ESG Issue	Performance Indicator		2022	2023	Reference
Environmental					
GHG Emissions & Climate Change	Direct GHG emissions (tCO ₂ e)		46,474	47,753	SASB EM-MM-110a.1 / GRI 305-1
	Indirect GHG emissions (tCO ₂ e)		12,376	11,644	GRI 305-2
	Direct and indirect GHG intensity (ratio)		FeNb = 1.61 Talc = 0.13	FeNb = 1.31 Talc = 0.16	GRI 305-4
Water & Effluents Management	Total water withdrawal (megaliters)		7,635	7,761	SASB EM-MM-140a.1 / GRI 303-3
	Total water used in operations (megaliters)		25,184	24,321	SASB EM-MM-140a.1 / GRI 303-5
	Total water discharge (megaliters)		8,709	8,466	GRI 303-4
	Percentage of water used that is recycled (%)		70	68	
	Water-related notices of violation (#)		1	1	
	Fresh water withdrawn in regions with High or Extremely High Baseline Water Stress (%)		0	0	
	Fresh water consumed in regions with High or Extremely High Baseline Water Stress (%)		0	0	GRI 303-4
Waste Management & Reduction	Total non-hazardous waste (t)		5,018	3,495	
	Total weight of hazardous waste generated (t)		217	285	SASB EM-MM-150a.7
	Total amount of overburden, rock, tailings, and sludges and their associated risks (t)		5,802,013	6,259,074	GRI MM3
	Total weight of tailings produced (t)		2,726,772	2,583,084	SASB EM-MM-150a.5
	Total weight of waste rock and overburden produced (t)		3,075,241	3,675,990	
	Total waste (t)		5,807,248	6,262,854	
	Non-hazardous wastes diverted from landfill (%)		43.5	46.8	
	Total weight of hazardous waste valorized (%)		49.3	41	GRI-306-5
	Total weight of hazardous waste recycled (%)		19.5	8	SASB EM-MM-150a.8
	Total weight of hazardous waste diverted from landfill (%)		68.8	49	
	Tailings used for other purposes, including paste-backfill (%)		56.3	57.5	SASB EM-MM-150a.5

ESG Issue	Performance Indicator		2022	2023	Reference
Social					
Employee Health, Safety & Well-being	Employee total recordable injury rate (TRIR)	Yes	4.23	1.79	SASB EM-MM-320a.1
	Contractor total recordable injury rate (TRIR)	Yes	2.81	3.67	SASB EM-MM-320a.1
	Overall total recordable injury rate (TRIR)		3.97	2.85	
	Near miss frequency rate (NMFR)	Yes	1.98*	2.06	SASB EM-MM-320a.1
	Fatalities (#)	Yes	0	0	SASB EM-MM-320a.1
Diversity, Equity & Inclusion	Total employees (#)	Yes	848	856	GRI 405-1
	Total female employees (%)	Yes	15	14	GRI 405-1
	Total female employees (#)		125	124	GRI 405-1
	Total leadership level employees		160	178	
	Leadership level female employees (#)	Yes	36†	39	GRI 405-1
	Leadership level female employees (%)		23	22	
	Employees under 30 (%)		14	12	GRI 405-1
	Employees 30-50 (%)		57	55	GRI 405-1
	Employees 50+ (%)		29	33	GRI 405-1
	Board members who self-identify as visible minorities (#)	Yes			GRI 405-1
Governance					
Responsible Supply Chain	Spending on local* suppliers (% of procurement budget)		37	43	GRI 204-1
Business Ethics & Transparency	Reports of alleged violations of our Code of Conduct (#)		2	2	
	Reports on Code of Conduct violations with substantiated misconduct (#)		0	0	
	Production in countries that have the 20 lowest rankings in Transparency International's Corruption Perception Index (t)		N/A	0	
Environmental & Social Management	Total non-compliance fines (#)	Yes	1	1	GRI 307-1
Financial Contributions	Total economic value generated (\$ US ('000))		407,990	426,597	GRI 201-1
	Employee wages and benefits (\$ US ('000))		99,653	102,589	GRI 201-1
	Payments to government (\$ US ('000))		9,100	7,287	GRI 201-1

* 2022 NMFR has been restated (from 1.25) to reflect the subsequent harmonization of our tracking methodology across our businesses for this KPI.
† Restated (from 45) to reflect correction of a data error.

Appendix – More About Magris

Niobium

What is Niobium?

Niobium is a light grey, crystalline, and ductile transition metal with an atomic number of 41. Natural Resources Canada, the US Department of Energy, and the European Commission have each classified niobium as a critical element or raw material, including for modern or emerging technologies.

Where Do We Produce Niobium?

We produce niobium in the form of standard grade FeNb that is composed of approximately 66% niobium with iron at our vertically-integrated Niobec mining, processing, and metallurgical facility located in the Saguenay-Lac-Saint-Jean region of Quebec, approximately 200 kilometres north of Quebec City.

Niobec is one of three primary niobium producers in the world and the only one located outside of Brazil. It has been operating continuously since 1976 and supplies approximately 10% of the global niobium market, with a production capacity of almost 12,000 t of FeNb per year.

In 2023, we employed over 542 people in full-time positions at Niobec, about two-thirds of whom were unionized.

Niobec has estimated proven and probable mineral reserves of 76.9 million tonnes. These mineral reserves should enable Niobec to operate until 2052 under its current mine plan. Niobec has a historical record of maintaining or increasing its mineral reserves through its exploration program, which leads us to believe that there remains significant potential to increase Niobec’s mineral reserves and extend its operational life beyond the current horizon.

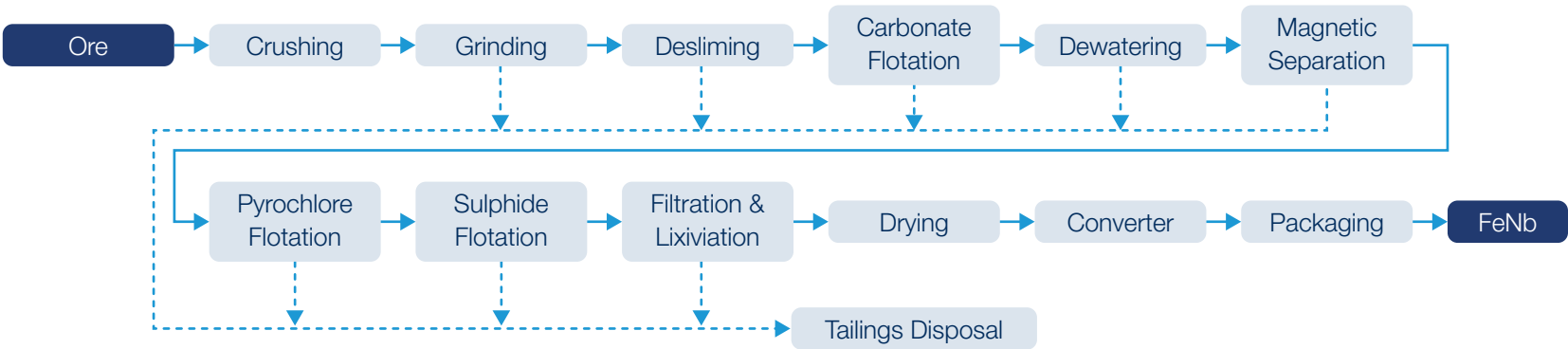
How Do We Produce Niobium?

Mining: At Niobec, we mine niobium-containing material (ore) underground using the open stope method, with an average-sized stope that is 45 metres in length, 24 metres in width, and 90 metres in height. Pillars that are 24 metres in width provide support between two open (emptied) stopes.

To extract the ore from a stope, we drill into the stope and load the holes with explosives which are then detonated. We gather and transport the blasted ore via an ore pass leading to a crusher where the ore undergoes primary crushing. We then hoist the crushed ore to the surface.

Processing: Once on the surface, we process the ore into a finer material in Niobec’s mill by further crushing, grinding, and sorting. We extract the niobium-bearing minerals using a complex process that involves desliming, magnetic separation, several flotation circuits, and leaching to remove unwanted waste constituents (tailings). We then dry the remaining material to produce a niobium oxide concentrate (Nb₂O₅). Over half of our tailings—which are largely chemically inert and of low risk to the environment—are sent to Niobec’s paste-backfill plant, where we transform the tailings into a paste that we use to fill in empty stopes, or as dam construction material for the tailings storage facility. Any remaining tailings are stored in the facility’s ponds.

Converting: We transfer the Nb₂O₅ concentrate in batches to a converter, where we use a proprietary aluminothermic reaction process that we tailor on a batch-by-batch basis to produce ferroniobium ingots containing specified amounts of niobium. After cooling, we store the waste slag produced in the process underground. We sample each ferroniobium ingot to confirm its chemical composition and then store it. We select and withdraw ingots from storage by matching their chemical content with customer requirements. We then crush the ingots in three size ranges, based on customer product size requirements, pack the final product, and transport it to customers by truck, rail, and/or ocean vessel. Our process to produce ferroniobium is illustrated below:



Key Niobium Production Considerations

Niobec is subject to a wide range of regulations concerning its environmental and overall health, safety, and well-being performance.

Environment

The principal federal environmental requirements governing our activities at Niobec are contained in the *Environmental Protection Act* (Canada), the *Fisheries Act* (Canada), and related *Metal and Diamond Mining Effluent Regulations* (MDMER) administered by Environment Canada.

The principal provincial environmental requirements to which Niobec is subject are contained in the *Environmental Quality Act* (Quebec) which is administered by the Quebec Ministry of the Environment’s Fight Against Climate Change in Wildlife and Parks (MELCCFP). Given Niobec’s focus on water management, we place great emphasis on compliance with Quebec Directive 019, which is administered by the MELCCFP and sets out requirements for the quantity and quality of discharge effluents and water for facilities with mining operations.

Niobec’s niobium recovery process requires significant quantities of water. As a result, a key focus at Niobec is compliance with the *Fisheries Act* and the MDMER, which are designed to protect fish, fish habitat, and fish use from effects in receiving waters due to the release of final effluents, and with Directive 019.

Integrated Chloride Management

Chloride in the form of sodium chloride (NaCl) occurs naturally in the Niobec ore body. A critical aspect of operations at Niobec involves how we manage chlorides in processing ore into our FeNb product since we expect NaCl concentrations will increase as we mine deeper into the ore body. Our management of chlorides is also complicated by our use of hydrochloric acid (HCl) to produce niobium concentrate during ore processing.

To address the ongoing challenge of chlorides management, Niobec has invested \$33 million in a water treatment plant to treat non-recycled water via high-efficiency reverse osmosis and evaporation and to ensure final effluent is fully compliant with all applicable water quality and quantity standards.

The water treatment plant investment, together with a leachate neutralization circuit, is part of an integrated chloride management system that treats the chemical constituents in water before it becomes final effluent.

Managing chlorides at Niobec involves several integrated actions: reducing our use of water containing chlorides and other chloride sources, neutralizing HCl used in processing ore, treating water to remove or reduce chlorides that could adversely impact the receiving water body for final effluent, and safely sequestering chloride-containing materials that result from our other actions.

Our water and chloride reduction actions involve a commitment to use the lowest possible amounts of water in our underground mining. One way we do this is by altering our water use practices to minimize water loss in the system. We have also adopted chloride-free alternatives where feasible, and completely or almost completely eliminated the use of chloride-containing products (e.g., winter de-icing salt).

As part of our integrated chloride management at Niobec, we also improved the paste-backfill plant, which turns tailings into a paste that used to fill in mined-out stopes in the mine and reduce the volume of tailings stored in Niobec’s tailings storage facility. These improvements enable the plant to use the chloride-containing brines in the paste it produces, allowing us to safely sequester the chlorides underground.

Niobec’s management of pluvial effluent (run-off) quantity and quality is also key to its ability to operate sustainably over the long-term. Water management at Niobec has been historically complicated by differing expectations of the federal and provincial regulators regarding how Niobec should manage its pluvial effluent. More recently, this has been further complicated by increasingly volatile water levels related to increased frequency and intensity of water events. By maintaining open and transparent communications with both regulators, Niobec was able to resolve these differences and is investing in pumping infrastructure that will manage the pluvial effluent. We expect to commission this pumping infrastructure by the end of 2024. In the meantime, Niobec is using an interim pumping installation to pump pluvial effluent within its existing water management system.

Workplace Health, Safety and Well-Being

Health, safety, and well-being in addition to other social aspects of Niobec’s operation are regulated by the Quebec *Commission des normes, de l’équité, de la santé et de la sécurité au travail* (CNESST).

Other Production Considerations

In addition to water management, tailings management is a key operational consideration for Niobec. While its tailings are considered “low risk”, since they do not contain potentially acid-generating constituents, the tailings must still be managed and safely stored. Niobec minimizes the quantity of tailings contained in its tailings storage facility by using tailings as much as possible for underground paste-backfill and as material for tailings dam construction and improvement.

Mine Closure and Reclamation

Niobec’s obligations for mine closure and reclamation are regulated under the *Mining Act* (Quebec) which also sets out requirements for other operational activities and the provincial regime for mineral rights ownership. Under the Act, Niobec must also periodically submit mine closure/reclamation plans for regulatory approval and provide financial assurance to the Quebec Ministry of Natural Resources and Wildlife for the cost of the closure and reclamation obligations in approved plans. The most recent closure plan’s cost is C\$13 million and an update is currently under review by the relevant authorities.

Key Operational Highlights

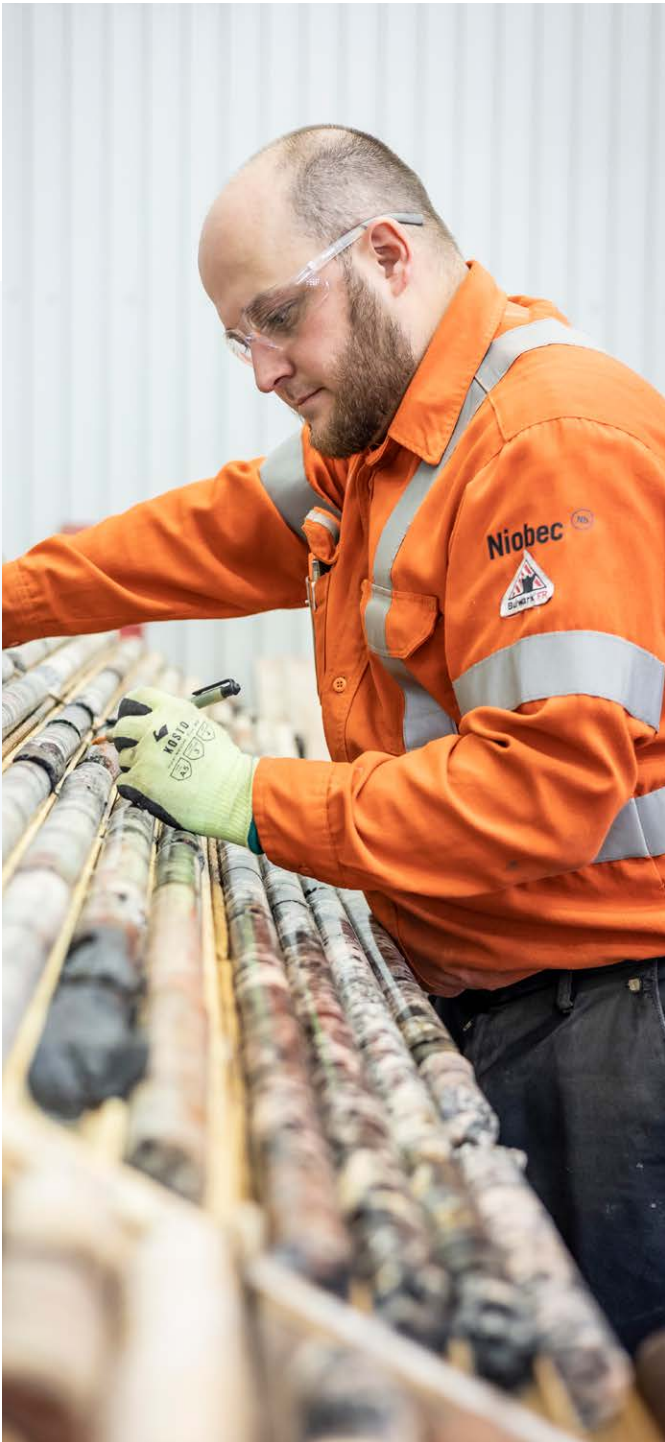
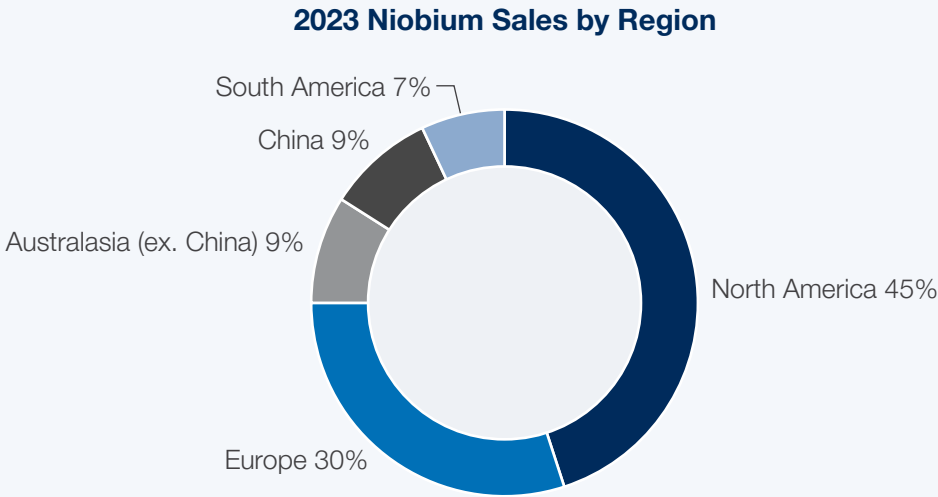
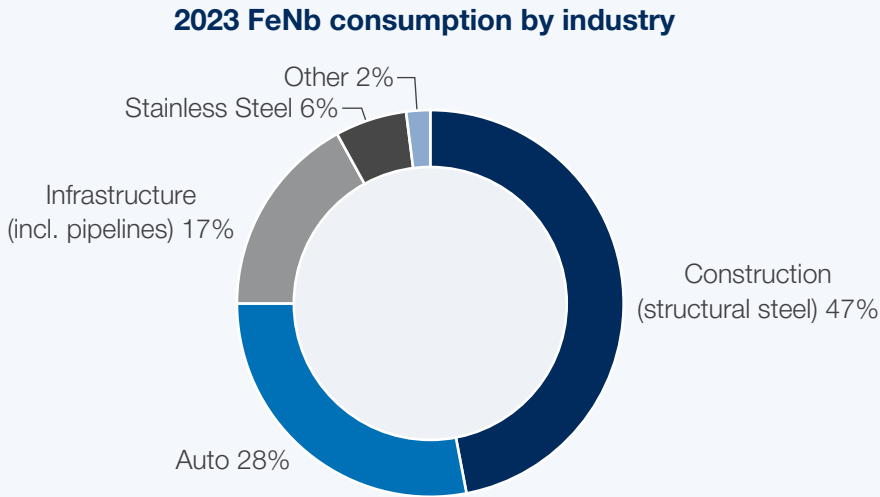
Niobec has been in operation for almost 50 years and has well-established relationships within the local community based on open and honest dialogue. Ninety-nine percent of the power Niobec purchases is generated from renewable sources and Niobec recycles approximately 80% of its water. Niobec has received ISO 14001 (environmental management system) and ISO 9001 (quality management system) certification since 1998 and 2009, respectively. It also currently holds a Silver sustainability rating from Ecovadis, a universally recognized sustainability ratings provider.

Our Niobium Product—Applications and Benefits

Our FeNb is a critical alloying agent used directly by steel producers to improve their grades of steel by improving strength, melting point, durability, and resistance to corrosion, so they can produce HSLA steel products. Even a small amount of niobium can significantly improve these properties in HSLA steels, reducing the amount of steel needed and increasing its durability in various applications, resulting in reduced energy and resource usage throughout our FeNb product life cycle.

Our Niobium Markets

The steel industry is global and so we sell our FeNb product to steel producers around the world. Niobium is not a commodity that is traded on a terminal exchange such as the London Metals Exchange, unlike other minerals such as copper, zinc, and nickel. With no terminal market available, we sell most of our FeNb product under bilaterally negotiated purchase and sale contracts with our customers, with some sales into the niobium spot market. 2023 global consumption of FeNb by industry is shown below:



Talc

What Is Talc?

Talc is a naturally occurring mineral with a platelet structure. It is the softest known mineral and relatively inert. Talc ores can differ widely in mineralogical and chemical composition as well as in purity and colour. A principal point of difference is platelet size. Talcs can have larger (macro) or smaller (micro) crystal sizes, which is an important factor in determining their specific properties.

Where is Talc Produced?

Portions of our Magris Talc business have a lengthy operating history, namely the Yellowstone mine which commenced production in the 1940s. We acquired the assets that comprise Magris Talc in 2021 from US and Canadian subsidiaries of Imerys S.A. under a Chapter 11 bankruptcy initiated by Imerys. Because the sale was completed under a bankruptcy process, we acquired the talc assets “free and clear” of all liabilities and claims associated with the period before our acquisition, including any litigation connected to the period prior to our ownership. Upon acquiring the talc business, we ceased production of talc for personal care products. With the acquisition, we became one of what are currently two fully integrated producers of talc in North America. Magris Talc is the leading talc producer in North America, supplying approximately 50% of the market. We are the only North American talc producer that can internally source both macro-crystalline and micro-crystalline ores.

We own and operate three talc mines, all of which are open-pit. Each mine has talc with distinctive properties that enable us to produce a wide range of talc products.

Name:	Yellowstone	Argonaut	Penhorwood
Location:	Cameron, Montana	Ludlow, Vermont	Penhorwood, Ontario
Crystal Type:	Micro	Macro	Macro

We have six facilities where we process talc ore that we mine and imported ore we purchase into the talc products that we sell. Magris Talc employs a total of 308 people.

Health, safety, and well-being in addition to other social aspects of Magris Talc’s operations are regulated by the US state and Canadian provincial occupational safety and health authorities in the jurisdictions in which its facilities are located.

How Do We Produce Talc?

Beneficiation refers to any process by which an ore is improved, or benefitted, by the removal of the material in the ore that has no economic value. For talc, beneficiation is simpler than most other minerals.

At our Yellowstone operation, we extract ore through straightforward mining—drilling and blasting ore and transporting it to Yellowstone’s sorting plant. The high purity of Yellowstone’s ore means no chemical beneficiation is needed. Once sorted, we ship the ore to our Sappington, Three Forks, and Houston facilities for further processing into our talc products. At our Vermont operations, ore beneficiation involves dry separation as part of the production process. At Penhorwood, we use a flotation process to beneficiate ore. Higher purity beneficiated ore is sent to the Timmins mill while lower purity ore is sent to the Penhorwood mill. We conduct all mining activities at Yellowstone while we use third-party contractors for mining activities at Argonaut and Penhorwood.

At our mills, we further beneficiate our talc ores by primarily affecting their particle size distribution and other physical characteristics through micronization, air classification, surface treatment densification, and compaction.



Key Talc Production Considerations

Like Niobec, Magris Talc is subject to legal and regulatory requirements relating to environmental protection, water use and management, occupational health and safety, mineral rights ownership, mining activities, and mine closure and reclamation. Since Magris Talc operates in multiple jurisdictions, these requirements can vary from operation to operation.

A key consideration for Magris Talc is ensuring the purity of its product. Magris Talc makes extensive efforts in this regard, including a formal program to sample and test ore for contaminants with a focus on asbestos. If such testing indicates the presence in the sampled ore of contaminants above applicable testing limits, the material from which the testing sample was taken is set aside and treated as waste.

Key Operational Highlights

Magris Talc conducts a widespread research and development program to customize its talc products with the special attributes required by customers for use in a wide array of applications. Its facilities have a long-standing history within the communities where they operate and have the benefit of a small environmental footprint that results from most of the facilities using minimal water and producing little or no tailings in their operations. All Magris Talc facilities have been ISO 9001 certified since 2000. MTU currently has a Bronze and MTC has a Gold sustainability ranking from Ecovadis.

Our Talc Products—Applications

Due to the wide variability of ores, talc lends itself to many different uses and applications. We make over 500 SKUs of talc that differ based on a variety of factors including brightness, grind size, and morphology. We expend considerable effort on research and development to create new talc products that are “engineered mineral solutions” designed to meet our customers’ needs.

Our talc products are used in applications that include:

- **Agriculture**—our products are used to protectively coat seeds and animal feed and enhance the efficiencies of fertilizer;
- **Construction**—our products increase crack and shrinkage resistance, corrosion resistance, and barrier properties in addition to easing application in adhesives, caulks, sealants, and asphalt roofing;
- **Paints and coatings**—our products increase viscosity, structure, pigment spacing, and corrosion resistance in protective and marine gels, gel coat primers, cure-in-place pipe, and architectural coatings;
- **Paper**—our products can add brightness and are used to control pitch and “stickies” in the paper production process;
- **Polymers**—our products are used in polymers for the manufacture of plastics for food containers, bagging, wire and cable coverings, automotive manufacturing (interior and exterior components), and in other plastics applications to provide or enhance tensile strength, stiffness, rigidity, impact resistance, lightweighting, and flame-retardant qualities;
- **Rubber**—our products add dimensional and thermal stability, lubricity, dusting, and reinforcement properties to hoses, window seals, and tires; and
- **Specialties**—our products are used in emissions substrates (catalysts) and technical ceramics to enhance thermal expansion, porosity, and extrusion properties while providing greater density and stability in ceramics.

Our facilities process ores with different properties and individual product profiles that supply customer end-use applications:

- **Houston**—polymers and chewing gum;
- **Ludlow**—polymers, paints and coatings, and construction;
- **Penhorwood**—polymers;
- **Sappington and Three Forks**—paper, paints and coatings, rubber, and construction; and
- **Timmins**—specials/catalysts, polymers, and paints and coatings.

On acquiring our talc business, we stopped producing talc for use in personal care products. We focus exclusively on talc products for industrial applications, with a very small portion of our talc production directed towards pharmaceutical applications and chewing gum.

Our Talc Markets

Like niobium, we sell our talc products directly to our customers under individually negotiated purchase and sale contracts or customized sale orders. In many cases, we have a deep and long-standing relationship with the customer. For example, the average relationship length with our top 50 customers is about 15 years.

Transport costs make up a significant portion of the overall cost of talc products. As a result, the global talc market tends to be more regionally distinctive than the niobium market. Magris Talc has over 500 talc product customers. The geographical distribution of our talc sales by dollar value is illustrated below:

